

Superdiffusive dynamics and nested oscillations in visual search

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Declaration

I, *Jude Luca Metcalf*, declare that this thesis is submitted in partial fulfilment of the requirements for the conferral of the degree *Bachelor of Science (Honours)*, from the University of Sydney, is wholly my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualifications at any other academic institution.

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Abstract

Gaze dynamics during visual search exhibit rich spatiotemporal structure, providing a valuable window into the neural computations underlying attention and perception. Active vision is an adaptive process in which information gathering is continuously guided by previous observations, engaging distributed neural circuits across the brain. However, existing models of gaze have largely remained divided between biological and predictive descriptions, and between the large-scale dynamics of saccades and the fine-scale movements occurring during fixations. In this thesis, we unify these perspectives within a single physics-inspired model, treating gaze as both a complex dynamical system and an adaptive process of probabilistic computation. Building on the theory of fractional neural sampling (FNS), we understand gaze trajectories through their dynamical signatures which are here demonstrated: global superdiffusion across spatial scales and the nested oscillations of regular saccades which modulate fine spatial sampling within fixations. This framework offers a potential basis for non-invasive characterisation of neural function, with possible applications to disorders of attention and perception, including ADHD, schizophrenia, and autism.

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I would like to thank my supervisor, Prof. Pulin Gong, for offering such an engrossing research project, balancing exciting results with an emphasis on the *so-what* of it all, and providing guidance across every area of research in my Honours year and in my undergraduate studies. I would also like to thank my co-supervisor, Dr. Brendan Harris, for the endless help and guidance in the most important areas of research, and for introducing me to the Julia hype. I am thankful of my broader research group, for bringing the Madsen Honours office closer to Physics. Among the Physics Honours cohort, I'm particularly glad to be a part of a smaller number who almost always find themselves at the SiFA research office, or else on the short walk to Ralph's café. Finally, and most importantly, I cannot thank enough my family and friends, across Australia and Germany, who—no matter how hard I've made it—always give their unceasing support at any distance.

Statement of student contribution

The central idea of this project was devised by my supervisor, Prof. Pulin Gong, and myself. The theoreitcal framework of fractional neural sampling (FNS) was previously developed by my supervisor to characterise the rich spatiotemporal dynamics of cortical neural activity. Building upon this theoretical foundation, my supervisor proposed extending the FNS framework to model gaze dynamics. My supervisor also provided a deep understanding of the literature involved—spanning the fields of physics, neuroscience, and machine learning—as well as providing the collaboration with Dr. Pascal Fries, Tim Näher, and Yufeng Zhang, who were invaluable in providing the necessary data and experimental framework to evaluate this work. My co-supervisor Dr. Brendan Harris provided a rich understanding of FNS theory as concerns cortical activity. My co-supervisor also provided practical advice as to computational implementation and tools to be used. My co-supervior provided the basis for the code which structures the numerical solving of the FNS equation in the Julia programming language.

My original contributions to this work are as follows. Broadly, my conceptual additions to this project were in the addition of an informative kernel to couple information gathering to model dynamics, and the interpretation of the model to treat a range of relevant spatiotemporal scales. Additionally, considering and quantifying the explored dynamical landscape as connected to a dynamical Bayesian target-belief distribution was also introduced by myself. The prediction and subsequent rediscovery of the high-frequency ocular microtremor in the eye-tracking data was initiated by myself Specifically, I provided the careful analysis of kernel-driven rhythmicity, local harmonic oscillations, and Lévy superdiffusion in this system (Chapter 2). The truncated solution of the fractional Fokker-Planck equation in this system, and dynamical approximations to find the relevant theta- and gamma-timescales was done by myself. The writing of the scripts required to simulate and analyse solutions to the modelled system in the Julia programming language was also done by myself, and can be found in the `VisualFNS.jl` project. I devised the methodology to compare the predicted dynamical signatures of the model with the provided eye-tracking data (Chapter 3). This includes the q–order fractional moments calculated and compared between the model and the data, which I presented through interpretation of multiscale superdiffusion as compared to literature on strong anomalous diffusion. This includes the saccade detection from time series extracted from modelled solutions, which involved adapting several classical methods of saccade detection with several pre-processing steps as presented and documented in the `VisualFNS_Analysis.jl` project. The construction and methodology of interpreting and calculating informational metrics based on modelled trajectories was provided by myself (Chapter 4). I defined the quantification of efficiency, under and guidance of my supervisors, and provided the interpretation of each considered dynamical signature and compared each to generalised forms of probabilistic computation. The code for defining and calculating these metrics can be found in the `VisualData_Analysis.jl` project.

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Introduction

The brain is among the most complex systems in science, able to efficiently build a rich internal model of the world from a large quantity of perceptual data. Nowhere is this clearer than in vision. In humans, an estimated one in five cortical neurons is devoted to visual processing and control [1, 2], yet despite this computational power, the efficiency of information processed is extreme: for every 1 MB of information received by the retina, only about 40 bit are consciously processed [3–5]. The visual system must therefore be highly selective, maximising the available information by actively sampling the environment rather than passively receiving information. High-acuity vision is confined to the fovea, so to build a useful representation of the visible world, an observer must move their eyes, directing the fovea towards successive locations of interest. These rapid ballistic movements, called saccades, organise periods of relative stillness known as fixations during which the brain gathers visual information, and together they form a sequence of decisions about where to look next, made several times every second during searching [6, 7]. Each decision depends on what has already been seen, what is currently in view, and the goals of the observer. This process of decision-making informing further visual perception is called active vision.

This makes gaze a valuable observable for understanding the brain. Gaze provides a direct, low-dimensional, continuous, and non-invasive proxy for neural attention [8–16], which is often difficult to track otherwise. Visual search, in which an observer scans a scene for a target with a determined goal, is among the most demanding and informative settings for studying decision-making in the brain. The subject must balance an exploration–exploitation trade-off, deciding when to continue sampling promising regions and when to discover new ones, while avoiding wasteful re-exploration of depleted regions. This must be done efficiently and robustly within the brain, as the cost of neural computation is high and the brain must be trained to perform well across many tasks [17–21]. The resulting trajectories which are found to be most efficient are rich and structured, with characteristic spatial statistics [22–24], temporal correlations [25, 26], and preferred timescales [8, 9, 27]. Understanding the mechanisms behind them offers a window not only onto vision, but onto the general strategies the brain uses to explore, sample, and decide under uncertainty [28–34].

The rich structure of gaze trajectories now invites a suitably complex physical description. A challenge in approaching this problem has been in combining the near-optimal information-gathering behaviour of gaze with the variability in the particular trajectories taken. In adressing the information-gathering capacity of gaze there has been success in mapping paths which maximise information gathered [35–38], and in addressing the variability of gaze, there has been much success in modelling gaze trajectories through a stochastic process [7, 24]. In this thesis we provide a well-motivated theory in the validity of such a stochastic model which maps complex adaptive cortical activity as described by the theory of fractional neural sampling (FNS) [11, 33] onto the dynamics of gaze. Considering these dynamics abstractly in exploring an informative landscape, we capture the information gathering which drives search tasks generally. By constructing such a model, we are able to ask and propose answers to quantitative questions about gaze dynamics: what are the functional roles of dynamical patterns such as memory effects in heavy-tailed fluctuations, adaptive local search strategies, and regular timescales in improving performance during a visual search?

Existing work has approached gaze dynamics largely one scale at a time, both in terms of space and time. Here we present a model which can address gaze dynamics across all scales. Studies of large-scale saccades characterise their timing, amplitudes, and kinematics [6, 39–41]. At the scale of whole search trajectories, stochastic models describe global strategies with heavy-tailed, Lévy-like exploration [42–44], or else through global information-maximising strategies [35, 45–53]. The faster- and shorter spatiotemporal scales of fixational eye movements are then treated as a separate subject, through stochastic descriptions of their high-frequency noise statistics [7, 54, 55] or through their perceptual role in rapid information-gathering [39, 56], as well as the low-amplitude drifts during

fixation [41, 56–58]. The various timescales of attention more broadly, such as theta rhythmicity, form a further literature rooted in neural recordings and behaviour [8, 9, 59]. Each account is valuable on its own terms, but they are developed with different formalisms, different data, and often different assumptions about what the eye is doing.

Each of these approaches leaves a gap, both between biological realistic accounts and empirical accuracy, and between the fine and coarse scales of perception which should each be captured under a single theory. Saccades and fixations are not independent processes: the regularity, size, and duration of saccades are influenced strongly by information gathered during fixation [6, 22, 60]. Models built for a single scale cannot capture this coupling, and their parameters are fitted separately and carry no common physical interpretation. As such, one cannot simply combine such theories into a patchwork theory, as this will not accurately describe the neural control of gaze dynamics. The combined large- and small-scale movements are also rarely measured together, so there are few quantitative constraints on how a unified model should behave across scales [61]. As a result, we lack a single, biologically motivated model in which the heavy-tailed spatial statistics of search, the rhythmic timing of saccades, memory of explored regions, and the structured dynamics of fixation arise from the same underlying biologically-informed mechanism. As such, a unified mechanistic account of search dynamics, going beyond fixation prediction, remains largely unexplored. Such an account must reconcile the phenomena above while remaining biologically plausible, parsimonious, and capable of generating testable predictions about novel or unexplained aspects of visual search.

In this thesis, we begin in Chapter 1 by surveying the relevant literature in studying the rich dynamics of gaze. This includes introducing the broad patterns of gaze trajectories which identify eye movements: in particular their division into jump-like saccades and periods of information gathering called fixations. We then consider the study of diffusive stochastic models which inform this work, followed by the motivating theory of probabilistic computation and neural sampling as driving the information-gathering dynamics of gaze. We then present the model studied throughout this thesis in Chapter 2. We discuss the key dynamical properties present in the model which are found across a large range of spatiotemporal scales: two temporal scales of nested oscillation, and a global spatial superdiffusion. Chapter 3 compares these predictions against real human eye-tracking data, finding validation in the significant regularity of saccade timings, superdiffusion across all spatial scales, and the novel interpretation of a high frequency ocular tremor found in human gaze. In Chapter 4 we present the strength of this model as it provides an understanding of how these dynamical signatures provide insight into patterns of brain function through the map between visual sampling and cortical sampling. These same functional advantages also apply to general probabilistic sampling strategies, such that the efficiency found in visual search can be understood broadly in terms of both a complex dynamical system, and a probabilistic computational system.

Chapter 1: Survey of the Literature

In this thesis, we aim to unify the rich gaze dynamics found during active visual search under a single physics-inspired model, admitting a deeper understanding of neural computation. There is growing success in turning methods from statistical physics and nonlinear dynamics towards the area of computational neuroscience. In particular, the rich and complex spatiotemporal activity of neural circuits inspires similar analysis to unify the dynamics of gaze trajectories across spatiotemporal scales, and present a physics-inspired dynamical understanding of neural computation in search tasks.

In Section 1.1, we review the background for developing a scientific understanding of the dynamics of visual gaze; the data we present in this thesis, the unique spatiotemporal patterns of such data, and the previous methods used to model complex dynamics found in gaze trajectories are each described. In Section 1.2, we introduce the theoretical framework of stochastic modelling, Lévy motion, and the anomalous superdiffusion found elsewhere in complex dynamical systems. Section 1.3 then provides the link between stochastic dynamics and computation sampling, both in biological and abstract contexts; providing the key motivation for developing a mathematical understanding of a gaze trajectory as a proxy for dynamical attention in the brain.

1.1 Gaze trajectories

Healthy human gaze trajectories are well characterised by a small set of dynamical patterns, the challenge is in understanding their interaction, functional roles, and mechanistic origin. The clearest pattern of gaze trajectories is a division into long jumps in positional attention called saccades, between which are periods of relative stillness called fixations during which information is gathered [6, 44, 62, 63]. Eye-tracking devices—such as the Eyelink 1000 used in collecting data for this thesis as shown in Figure 1.1**d)**—provide the high-precision measurements to analyse and model the gaze trajectories through continuous time, both during saccade and fixation.

Where the dynamics of fixation or gaze have been considered, they are treated as separate subject matter to each other, and rarely treated together in single studies. Gaze fixations exhibit three distinct dynamical patterns: near-instant microsaccades within the same point of focus, slow ocular drift about a point of focus, and high frequency ocular micro tremor [7, 66–68]. There are a number of studies examining a rich physical model of fixational dynamics [7, 54, 69], as well as developing the role such dynamics have in gathering information efficiently [27, 41, 70–73].

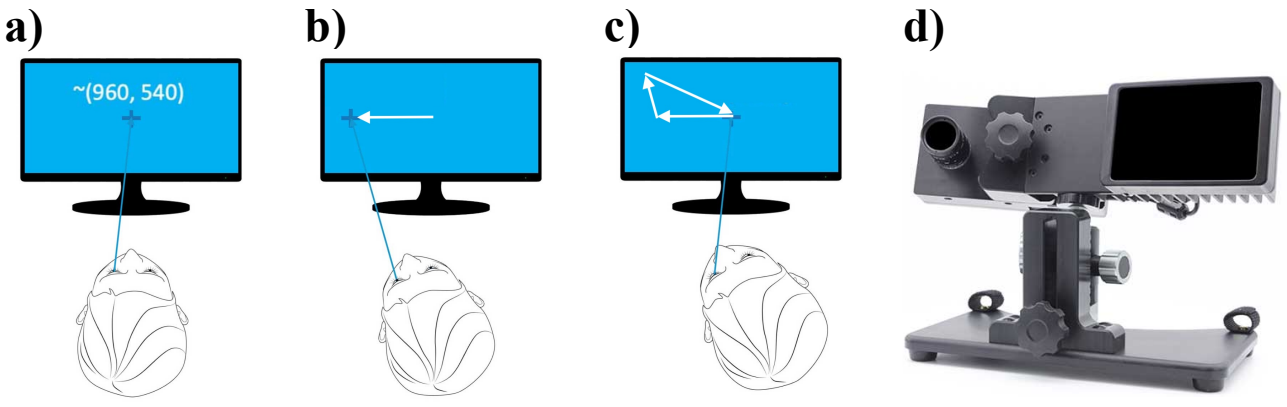


Figure 1.1: A schematic representation of a typical eye-tracking setup. a) shows a subject at the beginning of a typical eye-tracking study, with gaze position starting at the centre of the screen $(960, 540)$ px. **b)** shows a saccade from the centre towards the left of the screen as a target is tracked. **c)** shows the return of gaze position to the centre of the screen, as the target returns to the centre. **d)** shows a typical eye-tracking unit, the EyeLink 1000 Plus, as used in the experiment studied in this thesis. ***a)–c)*** *are adapted from [64], while* ***d)*** *is adapted from [65]*

1.1.1 Existing models of visual searching dynamics

Progress towards modelling gaze dynamics during visual search spans physics, neuroscience, and machine learning. As such, existing models have been built with very different aims and methods. Cognitive and biologically inspired models explain where attention is drawn through biologically-inspired saliency maps and top-down guidance [74–76]. More normative models ask what an optimally informative searcher would do, selecting saccades by Bayesian belief updating and entropy reduction [37]. We here examine in detail two influential models representing these two model-making paradigms, those being the self-avoiding random walk model of fixational dynamics presented in [7, 77], and DeepGaze III from [48] which utilises a deep learning model to predict saccade positions. A full comparison of a selection of influential and successful models, along with a comparison of their core domains, aims, and key features, is given in Table A.1 in Appendix A.

The self-avoiding random walk presented in [7] considers the dynamics within a single quadratic potential which confines the gaze into a localised fixation. The gaze position is discretised on a square lattice, and visited positions will increase potential to introduce a self-avoidance condition, which relaxes on a short timescale. Of particular interest to this thesis is the interpretation of the explored potential as a motor map from the neurons in the superior colliculus to motivate a physics-inspired model of gaze [78–80]. The self-avoidance condition, although taking a different motivation, is mechanistically very similar to the kernel adaptation which we introduce in this thesis to explain rhythmicity in global saccade timings. This work successfully reproduces superdiffusion in fixations, alongside the existence of microsaccades. This model, however, only accounts for fixational motion, and does not aim to capture the long saccades which organise them. Further, although the authors mention tremor, a tremor-frequency peak is not observed in this model.

DeepGaze III [48] produces fixation prediction through a state-of-the-art deep convolutional neural network, a map of observed oculomotor biases (such as an observed centre-bias in viewing an image), and a history of previous fixations, to predict the location of saccades with the highest accuracy to-date across all metrics considered in the standard MIT/Tübingen Saliency Benchmark [48, 81]. This model is highly successful in fixation prediction, and has been used to research patterns in attention through simulation of saccade prediction without requiring human measurement, however it lacks biological or functional usefulness beyond fixation position. The model provides no temporal information, only providing the location of fixations, and does not consider the dynamics of fixations themselves, nor providing any mapping from biologically realistic neural activity onto the dynamics of gaze.

1.1.2 Timescales of neural computation

Throughout much of this thesis, links are drawn between model dynamics and well-studied frequency bands of activity in the human brain. It is then valuable to outline the studied computational and functional interpretations of such frequency bands. Neural firing activity is divided into frequency bands named by greek letters, originally discovered and decomposed from electroencephalogram (EEG) data [82, 83]. Divisions are only approximate, but bands may be thus considered: delta (0.1–3 Hz), theta (4–8 Hz), alpha (8–14 Hz), beta (14–40 Hz), and gamma (40–100 Hz) [82, 83]. Of particular interest to this thesis are the interpretations of theta and gamma neural activity as functionally important to neural computation. Broadly, theta rhythms have been found to modulate and organise bursts in computational gamma-scale neural activity [9, 33, 84–87]. Theta rhythms have been shown to modulate attention during auditory tasks [14, 88], visual search tasks [8, 9, 89], and when analysing the broad attentional mechanism of the brain [59, 82, 83].

Theta-rhythmic cycles are finding strong interpretation in the cognitive neuroscience literature as organising and communicating information sampled from the environment for integration into the internal model of the brain [9, 84–87]. Strong direct evidence for this theory include correlations between intelligence and high theta peak power in EEG data during tasks requiring focus and attention [14, 59], and the discovered sensitivity of the visual system to react and search scenes on a theta rhythm [8, 9]. Accordingly, the firing patterns which define the lowest level of computation carried

out in the brain are regularly found to be gamma in timescale, with broadband signals being found within 40–100 Hz in a wide range of measurements of neural activity during cognitive tasks [90–94]. The coupling between the phase of theta activity and the amplitude of gamma activity is known as theta-gamma coupling (TGC). This phenomenon, a specific form of phase-amplitude coupling (PAC), is schematically represented in Figure 1.2. The evidence and theory behind TGC are described in a systematic review by Ursino and Pirazzini [92]. This thesis provides a computation view of the function of TGC the within the search dynamics of gaze as modulating periods of fixational attention on a slow theta-scale, during which information is dynamically sampled on a gamma scale.

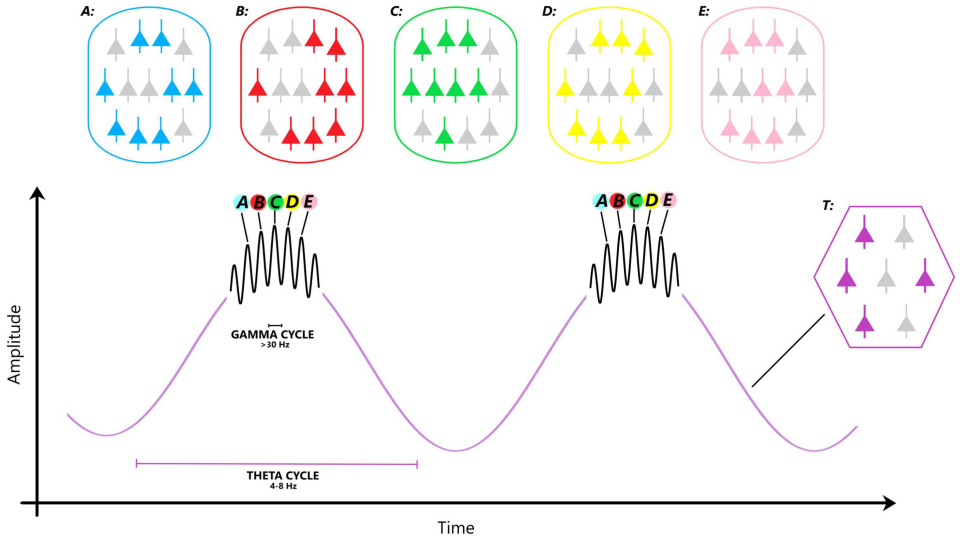


Figure 1.2: A schematic representation of theta-gamma coupling (TGC). Here, neural population patterns $\mathbf{A}-\mathbf{E}$ are activated in sequence on the timescale of a gamma oscillation $f > 30$Hz, triggered by the phase of a modulating theta wave of neuron population $\mathbf{T}$ on the theta timescale of $4-8$Hz. Such a formulation of TGC as correctly ordering signals has its foundations in the work of Lisman and Jensen [95], and was subsequently expanded to describe the dynamic neural mechanism during sensory exploration to modulate close attention upon individual centres of focus [96]. *Schematic adapted from Figure. 2 of Ursino and Pirazzini [92]*

1.2 Diffusive processes

In formalising empirical observations of human gaze and neural activity into a mathematical model, this thesis relies on the framework of non-equilibrium statistical mechanics and stochastic calculus. When a system is driven by a massive number of microscopic interactions—such as the complex interconnected firing of a neural population—modelling the exact deterministic dynamics becomes intractable. Instead, statistical physics provides the tools to coarse-grain these high-dimensional systems into low-dimensional stochastic differential equations. This section establishes the mathematical progression necessary to model scale-free visual inference. We first review the classical paradigm of normal diffusion and Brownian motion (Section 1.2.1), which serves as the foundational model for standard diffusive physical systems. We then examine how systems of interest frequently violate the classical assumptions of normal diffusion and Brownian motion, leading to an understanding of the presence and role of anomalous diffusion in many complex dynamical systems (Section 1.2.2). Finally, we introduce Lévy stable motion (Section 1.2.3), a class of superdiffusive fractional stochastic processes that mathematically capture the heavy-tailed, intermittent dynamics characteristic of optimal visual search and neural activity.

1.2.1 Brownian motion and normal diffusion

Statistical modelling has been fundamental in understanding complex dynamical systems: separating the dynamics of macroscopic observables into deterministic terms and stochastic terms reduces chaotic

high-dimensional systems to a small number of variables of interest [97, 98]. Stochastic processes are often the most natural, tractable, and accurate modelling tools when examining high-dimensional chaotic systems across physics, economics, geology, climate science, machine learning, and neuroscience [[**CITE**]]. The universality of these stochastic models is an implication of the central limit theorem: the distribution of the summed effect of a large number of identical and independently distributed (i.i.d) random variables of finite mean and variance will approach a Gaussian [99–101].

The most ubiquitous stochastic process is central to statistical physics: the Brownian motion W_t, equivalently called the Wiener process, which consists of stationary and independent Gaussian increments across all non-overlapping time intervals. A Brownian motion is defined only by its timescale, with a variance over trajectories after time t of t itself. Classical examples of processes modelled by the Brownian motion are: the movement of a macroscopic particle in a non-turbulent fluid [97], the gravitational influence of an orbiting collection of stars on supermassive black holes [102, 103], and (arguably) the movement of low-volatility stock prices [98]. The generalisation of the Brownian motion for heavy-tailed i.i.d variables is a Lévy flight, and is described in detail in Section 1.2.3. The comparison of these processes is shown in Figure 1.3.

Langevin dynamics of Brownian motion

Brownian motion of a variable $\mathbf{x}_t$ through time t about a potential landscape $V(\mathbf{x})$ can be described in terms of the first order Langevin equation:

$$d\mathbf{x}_t = -\gamma \nabla V(\mathbf{x}_t)dt + \sqrt{2\gamma}dW_t \tag{1.1}$$

where the constant γ describes the noise strength of the Brownian motion [104–107] and dW_t is a Brownian motion. Although the dynamics of $\mathbf{x}_t$ is described stochastically, the dynamics of the corresponding probability distribution $P(\mathbf{x}, t)$ is deterministic and can be described exactly. Such an equation of motion of $P(\mathbf{x}, t)$ under any stochastic dynamics defining $\mathbf{x}_t$ is called the Fokker-Planck equation [108–113]. In the case of Equation (1.1), the Fokker-Planck equation is [108, 109, 112]

$$\partial_t P(\mathbf{x}, t) = \gamma \nabla \cdot (P(\mathbf{x}, t)\nabla V(\mathbf{x}) + \nabla P(\mathbf{x}, t)) \tag{1.2}$$

whose unique stationary distribution is the Boltzmann-Gibbs distribution

$$\pi(\mathbf{x}) \propto \exp[-V(\mathbf{x})] \tag{1.3}$$

which yields the important relationship, where a Langevin potential can be constructed from a normalised target density via $V(\mathbf{x}) = -\ln \pi(\mathbf{x})$, which we will make use of in this thesis in constructing saliency maps from empirical data where only the likelihood of fixations $\pi(\mathbf{x})$) is given. Importantly, the Langevin system Fokker-Planck Equation (1.2) will describe the diffusion of a sharply peaked initial distribution as it widens, and the statistics of this diffusion from a sharp initial condition define normal diffusion as follows. The solution to the Fokker-Planck equation from an exact initial condition $P(\mathbf{x}, 0) = \delta^n(\mathbf{x} - \mathbf{x}_0)$ is dominated over short times by the ∇P term as $\nabla P \gg P\nabla V$, reducing the Fokker-Planck equation to the heat equation $\partial_t P = \gamma \nabla^2 P$. As such, the leading-order short-time solution is that of such a heat equation

$$P(\mathbf{x}, \Delta t) \propto \exp\left[-\frac{\|\mathbf{x} - \mathbf{x}_0\|^2}{4(\gamma \Delta t)}\right] \implies \mathrm{Var}[\mathbf{x}(\Delta t)] \propto \gamma \Delta t \tag{1.4}$$

which demonstrates the interesting behaviour which defines normal diffusion: the short-time scaling of variance in position will be linear, as $\mathrm{Var}[\mathbf{x}_{\Delta t}] = 2\gamma \Delta t$. Thus the diffusion coefficient γ can be thought of as the timescale of the Brownian motion, as it defines the rate of linear diffusion of variance with time. This linear scaling of the variance of trajectories with time defines normal diffusion [114–117], and understanding those systems whose fluctuations do not reproduce such normal diffusion is central in understanding stochastic motion complex dynamical systems.

1.2.2 Anomalous diffusion

The universality of normal diffusion in systems which are driven by Brownian motion as above has been studied closely, most influentially as in the movement of particles in a fluid [103, 114, 118, 119]. These dynamics inform the definition of normal diffusion through this variance scaling with time, rigorously calculated as the mean squared displacement (MSD) $\langle \|\Delta \mathbf{x}(\Delta t)\|^2 \rangle \propto \Delta t$. However, increasingly interesting physical systems have been found which diffuse nonlinearly in time, a situation known as anomalous diffusion, with an MSD scaling as:

$$\langle \|\Delta \mathbf{x}(\Delta t)\|^2 \rangle \propto \Delta t^{\lambda_2}, \quad \lambda_2 \neq 1 \tag{1.5}$$

Such anomalous diffusion yields two cases, subdiffusion $0 < \lambda_2 < 1$ where diffusion scales slower than normal, and superdiffusion $1 < \lambda_2 < 2$ where diffusion scales faster than normal. The case of $\lambda_2 = 2$ is known as ballistic diffusion, as it is found in the deterministic case of constant velocity $x = vt$.

Systems displaying anomalous diffusion are self-similar, and exhibit spatiotemporal scale-free statistics, as a result of their continuous-time power-law diffusive scaling. We will find in this thesis that it is advantageous in the study of anomalous superdiffusion to capture this multi-scale diffusive self-similarity through an extension of the λ_2–scaling of MSD to a spectrum of absolute moments $\lambda(q)$. This spectrum describes the diffusive self-similarity from the small spatial scales (low q) to large spatial scales (high q). Such a spectrum of absolute moments $\lambda(q)$ to capture diffusive scaling is [115]:

$$\langle \|\Delta \mathbf{x}(\Delta t)\|^q \rangle \propto \Delta t^{\lambda(q)}, \quad q > 0 \tag{1.6}$$

where q is the order of the fractional absolute moment, with $q = 2$ recovering the MSD, and $q = 1$ yielding a mean absolute displacement (MAD), and so on for any value $q > 0$. Systems exhibiting so-called strong anomalous diffusion (SAND) or multifractality will have a nonlinear spectrum $\lambda(q)$, meaning their self-similarity is differnt on large scales and small scales. A process which is uniformly self-similar on all scales will be defined by a linear spectrum $\lambda(q) = \frac{q}{\alpha}$. We will see that the process used in this thesis, a Lévy flight, is monoscaling (see Section 1.2.3 and Section 2.4).

Anomalous superdiffusion

Of great importance in studying the dynamics of visual search, the pattern of superdiffusive motion is found in many complex biological systems. We model the efficient exploration of gaze dynamics through Lévy motion, which is the standard dynamical model of superdiffusion in physics and is presented in detail in Section 1.2.3. Such Lévy superdiffusion has a strong link to the problem of designing a strategic search for scarce resources. The optimality of a superdiffusive process in a search has been reproduced in diverse functional search models [38, 120, 121], in a population of diverse searching agents with a range of superdiffusive exponents λ_2 [122] as well as in a large study against several different search conditions [120], and have been reported in the empirical search paths of biological organisms from the flights of drosophila flies [38, 123], albatross' searches [124, 125], and strongly across many marine predators [126, 127]. This thesis provides a biologically realistic and well-motivated superdiffusive search for research into a case for optimal Lévy motion. See Reynolds [128] for an in-depth discussion of a richer view of biological Lévy superdiffusion.

1.2.3 Lévy motion

To mathematically capture the intermittent, scale-free nature of visual search, we turn to the Lévy flight: a superdiffusive stochastic process that permits close analysis, due to it's very foundational mathematical definition. Here, the Lévy flight L_t^α is a stochastic process defined through it's increments being drawn from a heavy tailed symmetric α-stable distribution (a $\mathcal{S}\alpha\mathcal{S}$ distribution), where α is called the characteristic exponent and relates to the exponent of the distributions powerlaw tails. $\mathcal{S}\alpha\mathcal{S}$ distributions describe the limit of summation of i.i.d infinite-variance variables, as a result of the generalised central limit theorem, making them ideal to describe a universality class of superdiffusive processes. The α value defining a particular $\mathcal{S}\alpha\mathcal{S}$ distribution is a result of the asymptotic power-law

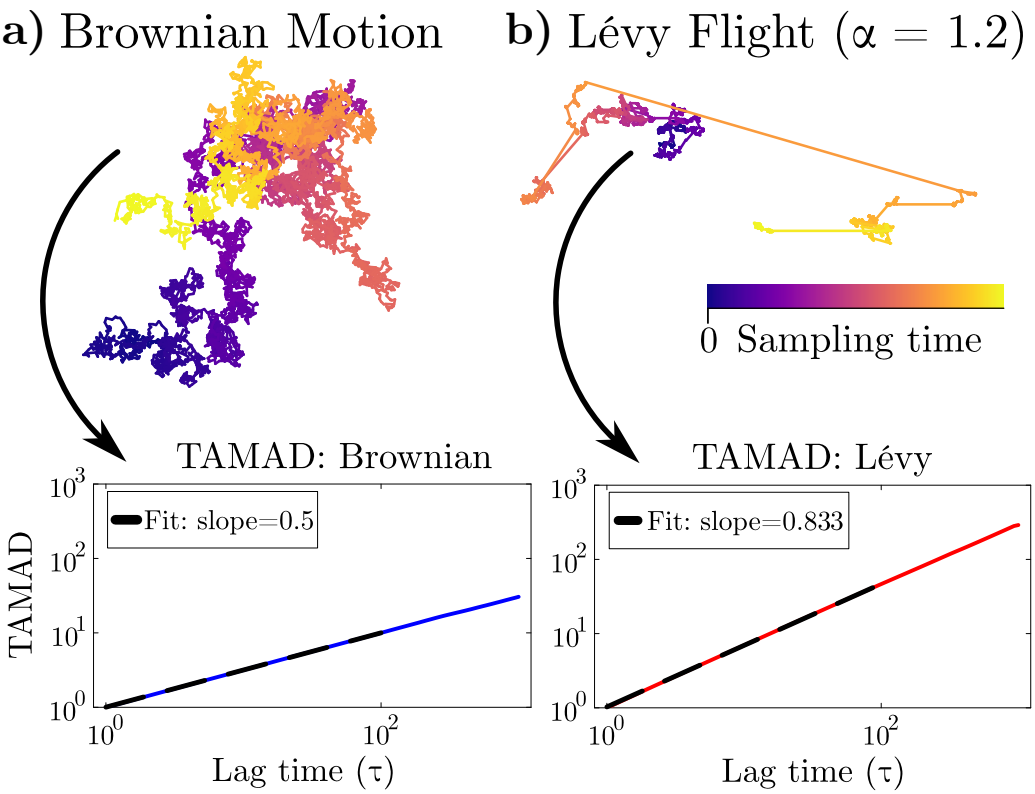


Figure 1.3: Schematic representations of a normally diffusive Brownian motion and a Superdiffusive Lévy flight. a) Schematic representation of a 2D Brownian motion, along with its time-averaged mean absolute displacement (TAMAD), which is the $q = 1$ mean fractional displacement as in Equation (1.6). Normal diffusion is here present as a power law slope of exponent 1/2, giving diffusive exponent $\lambda_1 = 1/2$ and $\alpha = 1/\lambda_1 = 2.0$. **b)** Schematic of a superdiffusive 2D Lévy flight with characteristic exponent of $\alpha = 1.2$, and a MAD slope of $\lambda_1 = 5/6 = 0.833\ldots$ corresponding to the theoretical superdiffusive coefficient $\alpha = 1/\lambda_1 = 1.2$.

tail of the density function, with the probability density function following $p(x) \propto |x|^{-1-\alpha}$. As a result, the Lévy flight L_t^α after time t is defined by its fractional scaling properties and alpha-stable statistics

$$L_t \overset{d}{=} t^{1/\alpha} L_1, \quad L_1 \sim \mathcal{S}\alpha\mathcal{S} \tag{1.7}$$

where $\overset{d}{=}$ denotes equal in distribution. In the limit $\alpha \to 2$, the Lévy flight reduces to a scaled Brownian Motion $L_t^{\alpha\to 2} = \frac{1}{\sqrt{2}} W_t$. If $1 < \alpha < 2$ the variance of L_t^α diverges and has mean 0, whereas if $0 < \alpha \leq 1$, then both the variance and mean diverge. In analysing the distribution of a Lévy flight, one must generalise the standard Langevin Fokker-Planck equation (1.2) to a Lévy motion using fractional calculus to capture nonlocal transport. To analyse this nonlocal behaviour of superdiffusion, the fractional Fokker-Planck equation is used, which yields the nonlocality, and scale-free behaviour of trajectories of Lévy flights (see the appendices to [33]). For an analysis of the fractional Fokker-Planck equation in this thesis, see Section 2.4. Many of the simple analytic properties of the Brownian motion are no longer true of a Lévy flight. Notably, since these flights have infinite variance in steps sizes, the mean squared displacement diverges. Thus their superdiffusion is better measured using an extension of MSD to fractional absolute displacements, through the definition of the moment spectrum function $\lambda(q)$ as in [115]:

$$\langle \|\Delta \mathbf{x}(\Delta t)\|^q \rangle \propto \Delta t^{\lambda(q)}, \quad \lambda(q) = \frac{q}{\alpha} \quad \alpha > q > 0 \tag{1.8}$$

where q is the order of the fractional absolute displacement, and the power law slope will converge for an α-Lévy flight to $\lambda(q) = q/\alpha$ when $q < \alpha$. As such, for Lévy flights with exponents $\alpha > 1$ superdiffusion

is well-characterised by the mean absolute displacement, where $q = 1$ in Equation (1.6).

Importantly, trajectories of a Lévy flight are no longer continuous. Instead, such trajectories are called *càdlàg* (continue à droite, limite à gauche) with respect to time, which describes functions which are right continuous with well-defined left limits [129]. Thus, we say that since all such trajectories produced from a Lévy flight are càdlàg, the Lévy flight itself is called a càdlàg process. Càdlàg processes have been widely studied in the context of mathematics, economics, and physics, and their properties are well understood [130, 131]. It has been shown that Lévy flights are semimartingales, and are well interpreted in integration [129]. Importantly, it follows from the Lévy-Itô decomposition theorem that a Lévy flight can be decomposed as the sum of two processes [132]:

1. A compound Poisson process of large jumps with $\|\Delta\mathbf{x}\| > 1$. Such jumps are exponentially distributed in time, and there are almost surely a finite number in any finite time period.
2. A square-integrable martingale (mean 0) jump process with $\|\Delta\mathbf{x}\| < 1$. This process is then finite-variance, and as such is subject to the central limit theorem, meaning that under temporal coarse graining it approaches a Brownian motion.

This division is useful in analysing the process in the context of nonlocal stochastic exploration in Lévy flights, which is modelled by the compound Poisson process of large jumps $|\Delta\mathbf{x}| > 1$, and the local exploitation dynamics modelled by the square-integrable martingale of small jumps $|\Delta\mathbf{x}| < 1$. In the context of visual searching, such a division naturally inspires the division between large jumps of attention, and periods of prolonged attention. As such, we are interested in why such superdiffusive scale-free dynamics may have a powerful in computational advantage in exploring an abstract unknown landscape.

1.3 Probabilistic computation

The stochastic dynamical exploration of an information landscape is a powerful paradigm of general computation, and is known as probabilistic computation. Such a landscape in this context is a representation of a target probability distribution. Sampling processes represent and evaluate these distributions through the efficient production of informative stochastic samples. Uncertain, high-dimensional, and ambiguous inference problems require efficient and accurate characterisations of target distributions. Sampling methods aim to draw sampled independently from a target distribution. This paradigm has become widely utilised in machine learning and statistics, where generative models learn patterns from sparse samples of large multivariate distributions inferred from observational data [31, 133]. This framework has recently gained significant traction as a basic scheme to describe biological computation. The variability in firing patterns of neural populations would thus maintain an internal model of perceptual uncertainty and evolving understanding, giving a functional role to chaotic neural activity [134–136]. This section outlines the paradigm of neural sampling as an interpretation of the variability in neuron firing patterns and visual search gaze trajectories (Section 1.3.1) as well as their extension to describe visual searching as a sampling process (Section 1.3.2).

1.3.1 Neural sampling

The framework of neural sampling posits this paradigm that the variability of cortical neural responses is an important functional feature of biological computation. Otherwise regarded as noise, stochastic neural activity patterns are thus presented as samples drawn from a posterior distribution over possible sensory interpretations [30, 137]. The neural sampling hypothesis has been shown to perform and predict network-level empirical patterns of neural activity in the V1, and recently by Terada and Toyoizumi in biologically realistic cortical networks, better than alternative theories [30, 32, 137, 138]. Perceptual uncertainty is then encoded by the variability of these cortical responses across time and across multiple trials, with higher uncertainty reflected in wider response distributions [30, 31, 137]. Neural sampling thus provides an interpretation for the way the brain navigates uncertainty at the fundamental scale of neural computation.

Studies into neural sampling show that variability is not strictly a result of explicit stochastic noise, as traditionally believed. Instead, recent work demonstrates that strongly recurrent, balanced inhibitory-excitatory deterministic neural networks can generate the high-dimensional variability which stochastic models are built to capture [32, 137]. Terada and Toyoizumi link this sampling of a posterior distribution to Bayesian learning through training a recurrent network to sample static features and dynamic trajectories, generalising learned priors to novel, uncertain environments through biologically plausible learning [32]. Further biologically plausible implementations of such Bayesian sampling are energy-based models (EBMs) which suggest that fast-spiking interneurons can estimate the local partition function to enable fast, localised sampling dynamics [34]. The sampling hypothesis has thus given a convincing theory of biological neural computation, with testable and well-supported evidence that neural variability has an important functional role in computation [30, 138]. Traditional Gaussian and Poisson noise models representing these stochastic network dynamics inherently correspond to continuous struggle to capture the full dynamics of complex interconnected inhibitory-excitatory (IE) biological neuronal network [139–141]. Further study into the dynamics of these IE networks has thus motivated further functional models of neural activity.

Fractional neural sampling

The FNS dynamics are called fractional, as the non-local dynamics of their distribution require fractional calculus to describe such spatial superdiffusion in Lévy flights [142]. The theory of FNS thus unifies the scale-free nonlocal dynamics of neural firing patterns with the neural sampling framework derived from the ideal Bayesian scheme of active inference. The critical advantage of FNS over other standard continuous random walks is its ability to efficiently sample complex, multimodal probability distributions with far-apart modes—likely a major feature of processed natural environments replete with multiple distinct salient features. While Brownian-motion-based samplers typically remain trapped in local modes due to high potential energy barriers, the heavy-tailed, large jumps inherent in FNS enable the neural sampler to adaptively and freely switch between multiple modes [143–146]. Furthermore, the generality of the FNS description permits the application of FNS equations of motion to gaze dynamics, as the motion of the eyes are driven by an IE neural circuit which is expected to keep key features of the FNS model invariant, such as superdiffusion, heavy-tailed statistics, and nonlocal interactions.

1.3.2 Visual sampling

This thesis presents visual search tasks as such a paradigm where a sampling model is well motivated, and driven by the underlying neural sampling process as well. The visual system is evolved to actively and efficiently process noisy information and evidence from the external world, and plan future search paths to gather further noisy evidence [28, 37, 147, 148]. In the context of visual searching, visually interesting targets with *saliency* as the equivalent of the target landscape, and where the information is located in a target image. Here, saliency is equivalent to the Boltzmann log-likelihood potential from the probability distribution of fixations $p(\mathbf{x})$

$$S(\mathbf{x}) = -\log(p(\mathbf{x})) \tag{1.9}$$

where visual search paths are most likely to be drawn to is a common technique whic [10]. As such, undirected visual searching of an image will show the areas most salient to the brain, allowing insight into the computational sampling mechanism of the visual system. In particular, search tasks require sophisticated integration of the cost and expected information gain of decisions, which invites fruitful decision analysis [149]. Empirical experiments show that eye movements show good agreement with greedy local information maximising fixations, given well-motivated visual sensitivity based on human physiology [28, 37, 150].

Chapter 2: A Unifying Model

The chapter is organised as follows. Section 2.1 defines the model based from FNS dynamics before presenting the search task used as the paradigm throughout this work. Section 2.2 introduces the dynamical-informative kernel which is novel to this work, giving its modelling motivations (Section 2.2.1) and the regular information-driven temporal structure of transitions between regions of interest as they are explored through informational kernel deposition (Section 2.2.3). Section 2.3 derives and quantifies the prediction from neural dynamics that localised trajectories produce high-frequency simple harmonic oscillations. Section 2.4 treats the defining characteristic of FNS dynamics, Lévy superdiffusion, deriving the short-time noise-driven scaling of fractional absolute moments and the monoscaling of self-similar superdiffusion which the model presents. Comparison of each of these dynamics with eye-tracking data is deferred to the next chapter.

2.1 Model setup

Mathematical formulation of gaze dynamics

The dynamical model of gaze dynamics, inspired by and developed from the FNS theory of neural dynamics (see Section 1.3.1, [33]), models eye movement trajectories as an overdamped Langevin system driven by an adaptive potential landscape $V_t(\cdot)$ by a kernel of information gathering $k(\cdot)$. Let $\mathbf{x}_t$ denote the two-dimensional position of visual attention, $\mathbf{p}_t$ a generalised auxiliary variable analogous to momentum and encodes short-scale memory effects, $\beta \geq 0$ the coupling strength parameter between $\mathbf{p}_t$ and $\mathbf{x}_t$, $\gamma > 0$ the noise strength parameter as in the standard Langevin system 1.1, and $0 < \alpha \leq 2$ the Lévy tail exponent which defines the heaviness of tails from which noise is drawn, and equivalently the strength of superdiffusion as in Sections 1.2.2 to 1.2.3. The driving noise process dL_t^α represents independent symmetric α-stable ($\mathcal{S}\alpha\mathcal{S}$) Lévy fluctuations.

A deterministic drift $b_\alpha(\mathbf{x}_t)$ drives the system driven by Lévy noise toward an invariant target distribution $\pi(\mathbf{x})$ over long times. As derived in [33], the exact non-local drift for arbitrary $0 < \alpha < 2$ is formulated in terms of the Riesz $(\alpha - 2)$–fractional derivative $\mathcal{D}_{\mathbf{x}}^{\alpha-2}$:

$$b_\alpha(\mathbf{x}) = \frac{\mathcal{D}_{\mathbf{x}}^{\alpha-2}[\pi(\mathbf{x})\nabla_{\mathbf{x}} \ln \pi(\mathbf{x})]}{\pi(\mathbf{x})}. \tag{2.1}$$

where these nonlocal fractional dynamics are treated in detail in Section 2.4. However, due to the computationally intense calculation of the nonlocal operator $\mathcal{D}^{\alpha-2}$, in this work we adopt the local gradient approximation of $\mathcal{D}_{\mathbf{x}}^{\alpha-2}$, which becomes exact as $\alpha \to 2$, as given in Equation (14) of [145]:

$$b_\alpha(\mathbf{x}) \approx c_\alpha \nabla_{\mathbf{x}} \ln \pi(\mathbf{x}), \qquad c_\alpha = \frac{\Gamma(\alpha - 1)}{\Gamma(\frac{\alpha}{2})^2}. \tag{2.2}$$

Combining these components, the full coupled system for gaze dynamics evaluated throughout this work is given by:

$$\begin{aligned} d\mathbf{x}_t &= -\gamma c_\alpha \nabla V_t(\mathbf{x})dt + \beta \mathbf{p}_t dt + \gamma^{1/\alpha} dL_t^\alpha, \\ d\mathbf{p}_t &= -\beta c_\alpha \nabla V_t(\mathbf{x})dt, \\ \partial_t V_t(\mathbf{x}) &= \frac{1}{\tau} k(\mathbf{x} - \mathbf{x}_t), \qquad V_0(\mathbf{x}) = S(\mathbf{x}) \end{aligned} \tag{2.3}$$

where as α, β, γ are parameters modulating FNS dynamics, c_α corrects the scale of the gradient drift, and τ controls information updates to the dynamical-informative potential $V_t(\mathbf{x})$. The nature and purpose of the dynamical updates $k(\cdot)$ is described in Section 2.2. Importantly, due to the equivalent relationship of a Langevin potential in to its targeted distribution in Equation (1.3) and the definition of saliency in visual sampling as in Equation (1.9), we can construct a continuous-time updating

internal-state belief distribution of the search target:

$$\pi_t(\mathbf{x}) \propto \exp\left[-V_t(\mathbf{x})\right] \tag{2.4}$$

Empirical protocol

To validate this modelled computational framework against human behaviour, we emulate the visual foraging task designed by Näher et al. [61], which requires subjects to sequentially scan an array of dots to locate a hidden target. This is done by choosing biologically accurate values for parameters $\alpha, \beta, \gamma, \tau$ and the information gathering scale of $k(\cdot)$. The task itself is encoded in the initial saliency map $V_0(\mathbf{x}) = \log \pi_0(\mathbf{x})$. These parameters and choice of $\pi_0(\mathbf{x})$ are described in the final part of this section.

In the human-testing protocol examined by Näher et al. [61], trials commenced with a pre-search fixation period of duration drawn uniformly between 2500–3500 ms. Following a 100 ms duration of no stimulus, participants were presented with a search array of white dots, each individual dot subtending 0.3° of visual angle on the screen. Exactly 20% of these dots functioned as true targets, which were visually identical to non-targets, and subjects were allotted a maximum search time of 5000 ms to successfully foveate a target, at which point an audio cue would indicate a success, and the subject would be required to fixate for another post-fixation period of 2500–3500 ms. To investigate the influence of spatial structure, the study compared a REGULAR task—where dots were spaced identically to the centers of an invisible hexagonal tiling—with an IRREGULAR task, which randomly redistributed the same number of dots across the screen to break spatial regularity. Assuming a standard viewing distance of 57.3 cm (where 1° ≈ 1 cm) and a monitor pixel density of 36.22 px cm^{-1}, spatial measurements translate directly between pixels and degrees using a scaling factor of 36.22 px $^{\circ -1}$.

Computational model & standard parameter setup

The empirical conditions from Näher et al. [61] are translated into numerical model parameters using spatial units of pixels (px) and temporal units of seconds (s). The baseline stimulus potential $S(\mathbf{x})$ is constructed on a 2D grid of Gaussian distributions representing quadratic potential wells over a 16×16 square grid (256 wells total), with 20% randomly designated as targets. Based on fixational eye movement distributions, the standard deviation of each well is set to $\sigma_{\text{well}} = 3$ px. The average axial distance between wells is fixed at 110 px (≈ 3.04° visual angle), bridging the dense (2.3°) and intermediate (4.5°) spacings in the REGULAR human search condition.

The baseline parameters for standard simulation runs are calibrated as follows:

- $\alpha = 1.2$: Captures the characteristic search superdiffusion observed in Section 3.3 while maintaining numerical stability.
- $\sigma_k = 80$ px: Defines an adaptive spatial acuity kernel by the Gaussian $k(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_k^2}\right)$. This matches the radial decay scale of human visual acuity (≈ 2.5°, corresponding to 87.2 px at a 50 cm viewing distance with 4 px mm^{-1} density).
- $\tau = 2 \times 10^{-4}$ s: Set to yield theta-rhythmic saccadic transitions, as derived in Equation (2.8) and evaluated in Section 3.1. (Note: across simulations, this produces an average ISI of 360 ms, mirroring the empirical saccade rate of 3.53 Hz / 280 ms average ISI).
- $\beta = 1000$ px s^{-1} and $\gamma = 1000$ px$^{\alpha}$ s^{-1}: Ensures recovery of gamma-rhythmic harmonic oscillations within the underdamped regime for 3 px wells (Section 3.2).

2.2 Dynamical-information adaptation

The most important component of the dynamics studied here is the dynamical-informative kernel. It is the principal extension of fractional neural sampling (FNS) work such as Chen and Gong [11] towards a full computational model of visual search during active vision. We call the kernel dynamical-

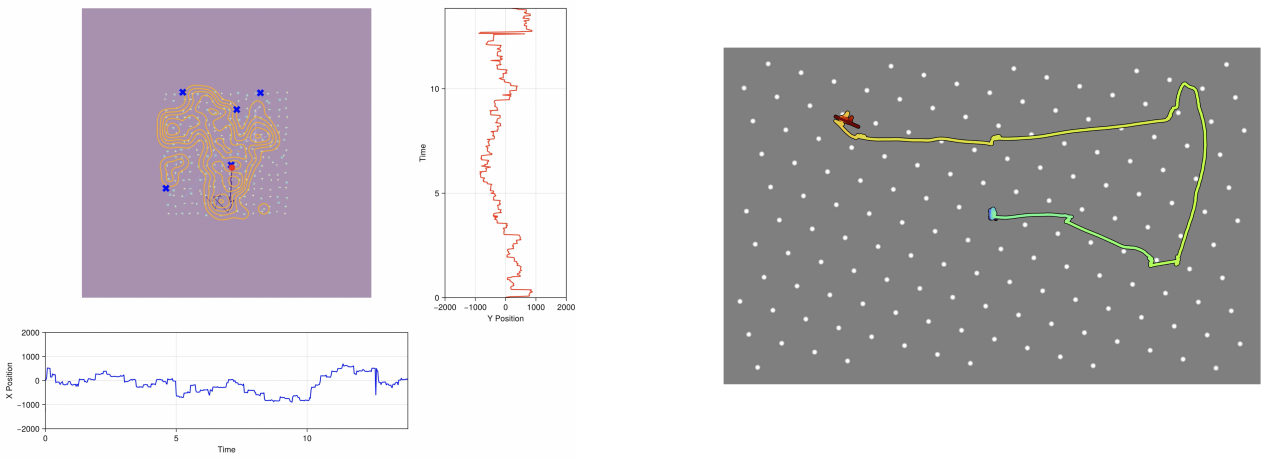


Figure 2.1: A comparison of the modelled search task with an example gaze trajectory from a real human trial.

informative because it couples two processes: the dynamics of the FNS sampler as it explores a landscape, and the information gained during that exploration when it is viewed as an active search. Its development therefore draws on two bodies of work. The first is statistical physics and computational sampling, where an analogous process of kernel deposition is called metadynamics (see Barducci, Bussi, and Parrinello [151] for its introduction and Bussi and Laio [152] for the "well-tempered" variant, which balances sampling accuracy against ergodicity). The second is the psychophysics of how spatial acuity limits information gathering during search.

The section proceeds as follows. Section 2.2.1 states what the kernel must achieve. Section 2.2.2 derives its form from Bayesian information gathering. **??** fixes its spatial scale from the eccentricity dependence of human visual acuity. Section 2.2.3 analyses the resulting dynamics and shows that they encode the theta-rhythmic timescale of saccades studied in Section 3.1, as well as the rate of information gain explored throughout Chapter 4. **??** collects the resulting parameters.

2.2.1 Motivations for kernel adaptation

The motivations for an addition of a dynamic landscape adaptation of the model are threefold. Firstly, the quantity of information gathered by the gaze trajectory must be accounted for, as this is the fundamental purpose of a visual search. Secondly, the regularity of information-seeking saccades during visual search must be present, most importantly the recently discovered phenomena of strongly theta-rhythmic saccades explored further in Section 3.1, as well as the well-studied phenomenon of inhibition of return (IoR) [40, 153]. Thirdly, any functionally informative description of search dynamics must couple these search dynamics to the current information which has been gathered complete the feedback loop of active vision [10, 53, 147, 154].

To couple gaze dynamics to information gained continuously, we examine the structure of visual information gathering. In this work the target distribution $\pi_t(\mathbf{x})$ will encode the belief in the presence a searched-for target object in 2D space, in a Bayesian sense. Thus, any information during exploration of non-target locations will cause a depletion of belief (probability mass) from the target distribution π locally. This is equivalent in the dynamical model to raising the potential landscape $V_t(\mathbf{x}) = -\ln \pi_t(\mathbf{x})$ locally by some kernel centred at the gaze position whose width represents the spatial acuity of information gathering. This gives a very well-motivated definition of continuous information gathering, as classical information is exactly equated to the logarithm of probability of 'surprise,' which is exactly $\pi_t(\mathbf{x})$ in our search task.

In this section we show that the following requirements are satisfied by the novel addition of a single

mechanism to FNS dynamics to construct a full model of gaze dynamics, the continual deposition of a kernel at the gaze position:

1. *Updating search information.* The quantity of information gathered along the gaze trajectory must be tracked, as this is the fundamental purpose of a visual search.
2. *Saccadic regularities.* The model must reproduce the regularities of information-seeking saccades: most importantly the recently discovered strongly theta-rhythmic saccades (Section 3.1), and the well-studied phenomenon of inhibition of return (IoR) [40, 153].
3. *coupling dynamics and searched information.* Any functionally informative description of search dynamics must couple those dynamics to the information gathered so far, completing the feedback loop of active vision [10, 53, 147, 154].

2.2.2 Developing the model kernel adaptation as information gathering

In this work the target distribution $\pi_t(\mathbf{x}) = e^{-V_t(\mathbf{x})}$ encodes, in a Bayesian sense, the belief that the searched-for target lies at position $\mathbf{x}$. Fixating a location $\mathbf{x}_0$ without finding the target is evidence against the target lying near $\mathbf{x}_0$, so considering some limited and spatially local information gathering during a fixation, belief must be depleted with according to some spatial kernel function $K(\cdot)$ as:

$$\pi_{t+\mathrm{d}t}(\mathbf{x}) \propto \pi_t(\mathbf{x})\, e^{-K(\mathbf{x}-\mathbf{x}_0)dt},$$

where $K(\mathbf{x}-\mathbf{x}_0) \geq 0$ is the log-evidence per unit time against discovering a target at $\mathbf{x}$ gathered during $\mathrm{d}t$, which is most informative at $\mathbf{x}_0$ where the most information may be gathered (see [35, 148]). The dynamics of $\mathbf{x}_t$ are then driven by the effective potential $V_t(\mathbf{x}) = -\ln \pi_t(\mathbf{x})$, so this multiplicative depletion of belief is an additive increment to the potential, $V_{t+dt}(\mathbf{x}) = V_t(\mathbf{x}) + K(\mathbf{x}-\mathbf{x}_0)dt$, up to a spatially uniform constant which normalises the distribution π_t and acts as a scalar gauge symmetry on the potential V_t.

Information gathering is therefore equivalent to raising the potential locally by the information gained at each location. This gives a well-motivated, continuous definition of information gathering, which turns out to be identical to an additive modulation of the potential surface which controls dynamics due to their shared link to an internal belief distribution π_t. The amount gathered at $\mathbf{x}$ depends on its displacement from the gaze position, because acuity falls off away from the fovea. Writing $K(\mathbf{x})dt = k(\mathbf{x}-\mathbf{x}_0)\,\mathrm{d}t/\tau$ to modulate the timescale use a dimensionless quantity $k(\cdot)$, the instantaneous law of kernel dynamics for gaze at $\mathbf{x}_t$ is

$$\partial_t V_{\mathrm{eff}}(\mathbf{x}) = \frac{1}{\tau}\, k(\mathbf{x}-\mathbf{x}_t), \tag{2.5}$$

where $k(\cdot)$ is an approximately radially symmetric function that tends to zero for large arguments and is scaled so that $k(\mathbf{0}) = 1$. Throughout this work we use an isotropic Gaussian,

$$k(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_k^2}\right). \tag{2.6}$$

the two parameters σ_k and τ control the spatial and temporal kernel dynamics respectively. The time constant τ is the time a gaze position needs to raise the potential by one nat at its centre, so $1/\tau$ is the peak rate of information acquisition. The width σ_k is the spatial scale of information gathering. We will see that τ is related to theta-rhythmic activity in gaze dynamics (Section 3.1). We can now show that σ_k is not a free parameter of our model but is set by the acuity of the human eye.

Relation of the kernel width to spatial acuity

The width σ_k is tied to how the detectability of the target falls with retinal eccentricity E, the angular distance from the point of gaze. This is quantified by the visibility map $d'(E)$, measured psychophysically for a given target and background, which ideal-searcher models [10, 35, 37, 147, 155]

use to weight the evidence from each fixation in the Bayesian update of target belief [36, 37]. The visibility map declines with eccentricity because due to the decreased resolution outside the central fovea [3–5], through the fall-off of acuity, cortical magnification and crowding, with a half-resolution eccentricity E_2 of a few degrees (about 2.3° in Geisler and Perry [156]) [157, 158]. For an observer with sensitivity d', the expected log-evidence against a target at a target-absent location is $d'^2/2$ per fixation [36, 37], so the kernel is the normalised squared visibility map,

$$k(E) = \left[\frac{d'(E)}{d'(0)}\right]^2,$$

and σ_k is the width of this profile. Thus σ_k is determined by the acuity scaling of the eye and the spatial scale of the target rather than being a free parameter. Calculating the E_2 of our model as the characteristic scale of our Gaussian σ_k, we find a $E_2 = \sigma_k \div 36.22\,\mathrm{px}^{\circ -1} = (80 \div 36.22)^\circ \approx 2.20^\circ$, which is biologically accurate for such a search task as we consider.

2.2.3 Adaptive kernel dynamics as a driver of long-time rhythmic exploration

We now predict analytically the rate of excursions under these modelled dynamics in Equation (2.3) due to kernel deposition. The central idea to understand these kernel-driven transitions is to form a tractable transition condition: the effective potential well becomes concave-down at the gaze position. Such a well no longer localises the gaze position $\mathbf{x}_t$, and the Lévy-driven noise will then produce excursions to other modes.

We consider a single isotropic Gaussian mode of the target distribution, centred at zero with standard deviation σ_{fix}, and gaze position fixating within it. In terms of the radial coordinate r about the mode, the effective potential $V_t(r)$ therefore starts as the quadratic well

$$V_0(r) = -\ln \pi_0(\mathbf{x}) = \frac{r^2}{2\sigma_{\mathrm{fix}}^2}.$$

The choice of Gaussian loses very little generality, because a Taylor expansion will show that locally every mode is approximated to second order by a quadratic well—referred to as Laplace's approximation. We assume that the gaze position is localised near $r \approx 0$ and that $\sigma_k \gg \sigma_{\mathrm{fix}}$, so that the small noise-driven motion about the tight potential well with width σ_{fix} can be smoothed over, and kernel deposition can be assumed to be centred at $r = 0$. Excursions are then driven by the deterministic kernel dynamics alone. Lévy noise will introduce exponential-distributed waiting times between large excursions, but analysing these is beyond the scope of this work, and this will only add a slight broadening of the distribution in waiting times modulated by the ratio of noise strength to kernel time constant τ/γ. Under Equation (2.5) with the gaze at the mode centre,

$$\partial_t V_t(r) = \frac{1}{\tau} \exp\left(-\frac{r^2}{2\sigma_k^2}\right),$$

which integrates to

$$V_t(r) = \frac{r^2}{2\sigma_{\mathrm{fix}}^2} + \frac{t}{\tau} \exp\left(-\frac{r^2}{2\sigma_k^2}\right). \tag{2.7}$$

The excursion criterion here is that the radial second spatial derivative of the explored potential at the fixation point $r = 0$ becomes negative, leaving the gaze unconstrained. The second spatial derivative is

$$\frac{\partial^2 V_t(r)}{\partial r^2} = \frac{1}{\sigma_{\mathrm{fix}}^2} + \frac{t}{\tau\,\sigma_k^2}\left(\frac{r^2}{\sigma_k^2} - 1\right) \exp\left\{-\frac{r^2}{2\sigma_k^2}\right\},$$

and at $r = 0$ (where the Hessian is isotropic) it vanishes at the time

$$t_\theta = \tau \frac{\sigma_k^2}{\sigma_{\text{fix}}^2}. \tag{2.8}$$

For $t > t_\theta$ the origin is a local maximum of V_t, making this explored region very low interest and unlikely to be explored, which yields a dynamical origin of inhibition of return [40, 66, 153, 159–161]: the gaze is expelled from the location it has just exploited. Strictly, the criterion marks the onset of delocalisation; the excursion to another mode is then carried by the Lévy noise.

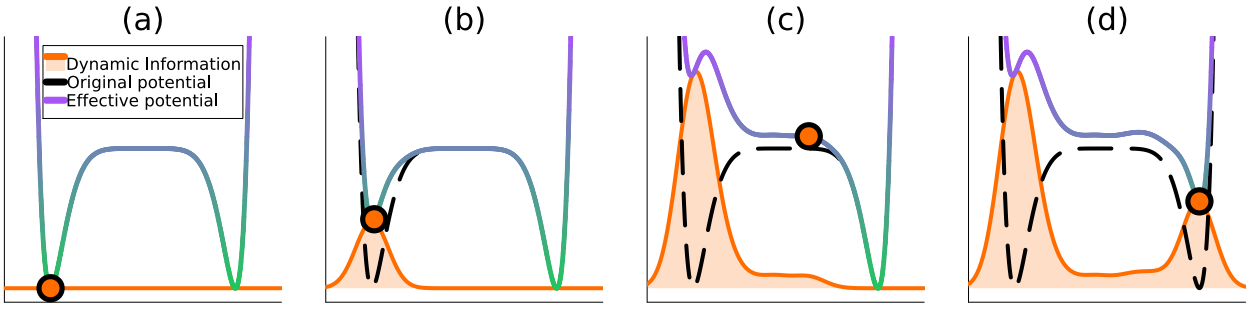


Figure 2.2: Schematic representation of kernel dynamics influencing an information-driven transition between potential wells. a) Two potential wells in a 1D potential landscape, with a high potential region separating them, which would represent low saliency in a model of gaze. $\mathbf{x}_0$ is positioned in the left well, which represents gaze position. **b)** Information acquisition as the kernel (orange) grows, gaining information and raising the potential landscape locally. **c)** A saccade-like transition, where the left potential well is inverted, followed by a transition to the right potential well. **d)** Exploration in the right well, with the information-gathering cycle resetting on a long t_θ-scale.

It is unsurprising that t_θ is proportional to τ. The dependence on the ratio of variances is of more interest. The dwell time is proportional to the variance σ_k^2 of the adaptive kernel and inversely proportional to the variance σ_{fix}^2 of the mode: a broad kernel takes longer to flatten a given well, and a sharply distributed fixation point will confine the gaze for longer. The result also extends beyond the Gaussian kernel, as σ_k represents the curvature of the kernel function at zero $k(\mathbf{0})$. An extension of the kernel method to functions which have a cusp at the origin may be biologically realistic [156–158], and could produce a rich description of the drifts and fixational movement of the eyes beyond tremor frequencies [41, 73], but such an approach is less tractable analytically and less stable computationally.

2.3 Local harmonic oscillations

One predictive feature of FNS dynamics is the local oscillations in sampling within confined potentials. We can quantify such high frequency oscillations under FNS dynamics by considering the deterministic part of our FNS equations of motion in Equation (2.3). Considering a Gaussian probability distribution of fixation positions $\pi(\mathbf{x}) \propto \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_{\text{fix}}^2}\right)$ and thus logarithmic gradients $\nabla \ln \pi(\mathbf{x}) = -\frac{\mathbf{x}}{\sigma_{\text{fix}}^2}$. Using this as our saliency landscape $\pi(\mathbf{x})$ without considering perturbation to the explored potential from the slowly updating kernel dynamics, our equations of motion are

$$\begin{aligned} d\mathbf{x}_t &= -\gamma c_\alpha \frac{\mathbf{x}_t}{\sigma_{\text{fix}}^2} dt + \beta \mathbf{p}_t dt + \gamma^{1/\alpha} dL_t^\alpha, \\ d\mathbf{p}_t &= -\beta c_\alpha \frac{\mathbf{x}_t}{\sigma_{\text{fix}}^2} dt \end{aligned} \tag{2.9}$$

where we ignore the stochastic noise driving this equation to only consider the coupled ODE which describes deterministic part of $\mathbf{x} = \mathbf{x}_t$ and $\mathbf{p} = \mathbf{p}_t$, where the loss of subscript in typical when moving

from an SDE to an ODE

$$\begin{aligned}\frac{d\mathbf{x}}{dt} &= -\gamma c_\alpha \frac{\mathbf{x}}{\sigma_{\text{fix}}^2} + \beta \mathbf{p},\\ \frac{d\mathbf{p}}{dt} &= -\beta c_\alpha \frac{\mathbf{x}}{\sigma_{\text{fix}}^2}\end{aligned} \tag{2.10}$$

Assuming an underdamped regime where the system is periodic, corresponding to parameter range $\beta > \gamma\sqrt{c_\alpha}/\sigma_{\text{fix}}$, the resonant ordinary frequency $f = 2\pi\omega$ is given by

$$f = \frac{1}{2\pi\sigma_{\text{fix}}}\sqrt{\beta^2 c_\alpha - \frac{\gamma^2 c_\alpha^2}{4\sigma_{\text{fix}}^2}} \tag{2.11}$$

2.4 Lévy superdiffusion

To model the superdiffusion of a computational exploratory dynamical system such as found in searching eye movements, our system is driven by Lévy noise withing an over-damped Langevin regime (see Sections 1.2.1 to 1.2.3). Such underlying dynamics are known as Lévy flight (see 1.2.3) and are the paradigm superdiffusive system. However, both the potential-driven adaptive kernel dynamics, and auxiliary momentum dynamics add complexity to the superdiffusive properties of the model. Quantifying superdiffusion in this system exploring a confined and dynamic potential landscape requires a dedicated analysis of how Lévy dynamics will introduce monoscaling superdiffusion which is self-similar on all scales, with the same superdiffusive patterns driving large jumps across a landscape as those small-scale jumps when considering finer movements. Lévy motion unifies these scales under a single superdiffusive pattern, motivating our unification of gaze dynamics under a single model.

Analysis of Lévy flight superdiffusion from the fractional Fokker-Planck equation

Lévy processes are the canonical form of monoscaling superdiffusion, yet the analysis of the precise form of diffusive scaling requires close analysis. Lévy flight superdiffusion is analysed from the deterministic dynamics of the probability distribution of jump size $P(\|\Delta\mathbf{x}\|)$ dynamics of the probability distribution. This problem of understanding the dynamics of the distribution of $\mathbf{x}$ is identical to that approached when studying classical Brownian motion where the classical Fokker-Planck equation was developed (see 1.2.1). Here in the case of a generalised Lévy-driven Langevin system with momentum coupling, the deterministic dynamics of the distribution of $\mathbf{x}$ is governed by the fractional Fokker Planck equation, as described in [33, 162, 163]. We can consider the phase space of a generalised coupled set of n stochastic differential equations of d-dimensions of $\mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2, \cdots, \mathbf{z}_n]^T$, where $d\mathbf{z}_t = (\mathbf{D}+\mathbf{Q})\, b(\mathbf{z}_t)dt + \mathbf{D}^{\frac{1}{\alpha}} d\mathbf{L}_t^\alpha$ for $(n\times d)$–dimensional Lévy noise. Here, $\mathbf{D}$ is the positive semi-definite matrix which defines strength and coupling of noise across $\mathbf{z}$, and $\mathbf{Q}$ represents the skew-symmetric matrix defining dynamical coupling across $\mathbf{z}$. For such a system the fractional Fokker-Planck equation can be expressed exactly:

$$\partial_t P(\mathbf{z}, t) = -(\mathbf{D} + \mathbf{Q})\nabla_{\mathbf{z}} \left[b(\mathbf{z})P(\mathbf{z}, t)\right] - \mathbf{D}\mathcal{D}_{\mathbf{z}}^\alpha P(\mathbf{z}, t) \tag{2.12}$$

where $\mathcal{D}_{\mathbf{z}}^\alpha$ represents the α-fractional Riesz derivative, which generalises the operation of the Laplacian to fractional orders, and is defined in Fourier space as $\mathcal{D}_{\mathbf{z}}^\alpha[f] = \mathcal{F}^{-1}\|\mathbf{k}\|^\alpha \mathcal{F}[f]$, with the case of $\alpha = 2$ restoring the ordinary Fokker-Planck equation with $\mathcal{D}_{\mathbf{z}}^{\alpha=2} = -\nabla_{\mathbf{z}}^2$. It can be shown that our model of gaze dynamics is recovered for $\mathbf{z} = [\mathbf{x}, \mathbf{p}]$, $\mathbf{D} = \begin{bmatrix}\gamma & 0\\ 0 & 0\end{bmatrix}$ and $\mathbf{Q} = \begin{bmatrix}0 & -\beta\\ \beta & 0\end{bmatrix}$, shown in [33]. The $b(\mathbf{z})$ drift term required to recover a given distribution is presented in Equations (2.1) to (2.2). Any solutions to the dynamics in Equation (2.12) are nontrivial, as the fractional derivative results in a nonlinear multivariate integro-differential equation which is difficult to propagate, even numerically.

When considering diffusion over short timescales, the physics of fractional Langevin systems with $\alpha > 1$ are noise dominated. In our Lévy driven system, short times are dominated by α-stable dL_t^α Lévy noise,

across power-law distributed spatial scales of short jumps and nonlocal jumps to distant positions in short times. This is corroborated by the scaling of Lévy noise as $dL^\alpha_{\Delta t} = \Delta t^{1/\alpha} dL^\alpha_1$ in Equation (1.7) is stronger than ballistic effects which scale in short times as Δt. Thus, considering the fractional Fokker-Planck equation evolving from a delta function initial condition $P(\mathbf{z},0) = \delta^n(\mathbf{x}-\mathbf{x}_0)\delta^n(\mathbf{p}-\mathbf{p}_0)$ evolving through a quadratic potential using the position coordinate $\Delta\mathbf{x} = \mathbf{x} - \mathbf{x}_0$, yielding a drift term $\mathbf{b}(\Delta\mathbf{x}) = \mathbf{b}(\mathbf{x}_0) + \mathcal{O}(\Delta\mathbf{x})$. We can construct a leading-order short time solution up to $t = \Delta t$ and marginalise over the auxiliary momentum $\mathbf{p}$ which in practise is a nuisance variable, we have:

$$\partial_t P(\Delta\mathbf{x}, t) = -\gamma \mathcal{D}^\alpha_{\mathbf{x}} P(\Delta\mathbf{x}, \Delta t) - \nabla_{\mathbf{x}} \cdot \left[(\gamma \mathbf{b}(\Delta\mathbf{x}) + \beta\, \mathbb{E}[\mathbf{p} \,|\, \Delta\mathbf{x}, \Delta t])\, P(\Delta\mathbf{x}, \Delta t) \right] \tag{2.13}$$

where we can truncate these terms to highest order using

$$\mathbf{b}(\Delta\mathbf{x}) = \mathbf{b}_0 + \mathcal{O}(\Delta\mathbf{x}), \qquad \mathbb{E}[\mathbf{p}|\Delta\mathbf{x}, \Delta t] = \mathbf{p}_0 + \mathcal{O}(\Delta t), \qquad \mathbf{v}_0 = \gamma\, \mathbf{b}_0 + \beta\, \mathbf{p}_0 \tag{2.14}$$

which gives

$$\begin{aligned}
\partial_t P(\Delta\mathbf{x}, \Delta t) &= -\gamma \mathcal{D}^\alpha_{\mathbf{x}} P - \nabla_{\mathbf{x}} \cdot \left[(\mathbf{v}_0 + \mathcal{O}(\Delta\mathbf{x}) + \mathcal{O}(\Delta t))\, P \right] \\
\partial_t P(\Delta\mathbf{x}, \Delta t) &= -\gamma \mathcal{D}^\alpha_{\mathbf{x}} P - \mathbf{v}_0 \cdot \nabla_{\mathbf{x}} P + \left[\mathcal{O}(\Delta t^0) + \mathcal{O}(\Delta t^{1-1/\alpha}) \right] \cdot P \\
\partial_t \hat{P}(\mathbf{k}, \Delta t) &= -\gamma \|\mathbf{k}\|^\alpha \hat{P} - i\, \mathbf{v}_0 \cdot \mathbf{k}\, \hat{P} + \mathcal{O}(\Delta t^0) \qquad (2.15) \\
\implies \hat{P}(\mathbf{k}, \Delta t) &= \exp\left[-\gamma \|\mathbf{k}\|^\alpha \Delta t - i \mathbf{v}_0 \cdot \mathbf{k}\, \Delta t \right] \left[1 + \mathcal{O}(\Delta t) \right] \qquad (2.16)
\end{aligned}$$

Showing that, to first order in Δt, the short-time diffusion of under our model Equation (2.3) yields a $\mathcal{S}\alpha\mathcal{S}$ distribution which is instantaneously transported with velocity $\mathbf{v}_0 = \gamma\mathbf{b}_0 + \beta\mathbf{p}_0$.

Quantification of Lévy superdiffusion monoscaling

Following on from the solution for the characteristic equation $\hat{P}(\mathbf{k}, \Delta t) = \exp\left[-\gamma \|\mathbf{k}\|^\alpha \Delta t - i \mathbf{v}_0 \cdot \mathbf{k}\, \Delta t \right] [1 + \mathcal{O}(\Delta t)]$, we reduce to the characteristic function of Fourier-space jump sized $k = \|\mathbf{k}\|$. For $0 < q < 2$ this change of variables will give the relation [164]

$$\langle \|\Delta\mathbf{x}\|^q \rangle = C_q \int_0^\infty \left[1 - \operatorname{Re} \hat{P}(k, \Delta t) \right] k^{-1-q}\, \mathrm{d}k, \qquad C_q = \frac{q(1-q)}{\Gamma(2-q)\cos(q\pi/2)}, \tag{2.17}$$

where C_q is defined at $q = 1$ by its limit, $C_1 = 2/\pi$. Substituting our result of Equation (2.16) into Equation (2.17) and evaluating after a further change of variable $u = \gamma\, \Delta t\, k^\alpha$, we have for $0 < q < \alpha$:

$$\langle \|\Delta\mathbf{x}\|^q \rangle = C_q \left[\frac{\Gamma\left(1 - \frac{q}{\alpha}\right)}{q} (\gamma \Delta t)^{q/\alpha} + \frac{\Gamma\left(\frac{2-q}{\alpha}\right)}{2\alpha} \frac{v_0^2}{\gamma^2} (\gamma \Delta t)^{(2-\frac{2}{\alpha})+\frac{q}{\alpha}} + \mathcal{O}\left(\Delta t^{1+q/\alpha} \right) \right]. \tag{2.18}$$

where the first term is the moment of an α-stable Lévy flight with scale $(\gamma\Delta t)^{1/\alpha}$, which is finite only for $q < \alpha$ [165]. The second term is the ballistic contribution of the momentum, which will tend to dominate as oscillations over longer times. Since $2 - \frac{2-q}{\alpha} > \frac{q}{\alpha}$ for $\alpha > 1$, it is subleading at short times, effecting dynamics more as $q \to 0$ and $\alpha \to 1$. This gives diffusion through fractional absolute moments scaling as

$$\langle \|\Delta\mathbf{x}\|^q \rangle \propto \Delta t^{\lambda(q)}, \qquad \lambda(q) = \frac{q}{\alpha}, \qquad 0 < q < \alpha. \tag{2.19}$$

Thus, for $q < \alpha$ the modelled dynamics of Equation (2.3) reproduce the scaling of a Lévy flight [146]. Such superdiffusion will produce a self-similar process with a single scaling exponent $1/\alpha > 1/2$ across all scales. This is called monoscaling superdiffusion, where superdiffusion is present and identical across all scales of q.

However, for $q \geq \alpha$ the moments of an unbounded Lévy flight diverge and the expansion above no

longer applies. It is thus important to computationally examine the behaviour of a Lévy flight during exploration of particular potentials, whether or not they are unbounded or dynamic, and if any post-processing is done to handle divergences in the variance of jumps. In other studies, Lévy flights have been found to have $\lambda(q) \neq \frac{q}{\alpha}$, and are thus proposed to be multifractal with different scaling behaviour for large jumps as for small. This is the signature of strong anomalous diffusion [166], however it is misleading, as a Lévy flight is defined by monoscaling self-similarity. In this study, we find that under the confined dynamics and light low pass filtering to produce continuous trajectories, the Lévy flight yields a single monoscaling superdiffusive exponent of $\lambda(q) = \frac{q}{\alpha}$

Summary

Throughout this chapter, we have described three key dynamical components of the unifying model of gaze dynamics we present: a kernel which couples regular information gathering with dynamical transitions, the high frequency oscillations which drive fine spatial dynamics, and superdiffusive spatial dynamics which are defined by a single exponent across all spatial scales. In Section 2.1 we compared the modelled setup with an empirical experiment performed in [61]. To examine this model as a valuable predictive and explanatory model of gaze dynamics, we compare to these human data.

Chapter 3: Empirical Validations

Our model is designed to be compared against empirical data and against established accounts of the phenomenon it describes, rich gaze dynamics during visual search. In this chapter, we compare each of the modelled dynamics of Chapter 2 with high precision eye-tracking data gathered by collaborators in Näher et al. [61] as described in Section 2.1. gaze dynamics during visual search show many broad and often task-dependent spatiotemporal patterns, so we ask which are most important for understanding computation in the brain. The literature inspires our partition of modelled dynamics, namely 2.2 information rhythmicity, 2.3 local short-time oscillatory sampling, and 2.4 global spatial superdiffusion, maps onto important visual dynamical signatures that have been observed elsewhere. These correspond to (1) theta-rhythmic macrosaccades, (2) gamma-scale 40–100 Hz fixational ocular drift, and (3) global saccadic superdiffusion to an informative visual scene [56, 167]. Importantly, our unifying model provides an understand of each of these dynamics as a function of network-level neural activity, including the novel connection between microtremor and the neural gamma signal, theta-gamma coupling, and the monoscaling of Lévy superdiffusion across spatial scales.

Section 3.1 examines theta rhythmicity in saccades, comparing the model's transitional rhythmicity with the saccade timing in the data. Section 3.2 examines gamma-scale oscillations during visual fixations, testing the model's local high-frequency harmonic oscillations against fixational drift. Section 3.3 examines superdiffusion in search, assessing whether the global spatial statistics of gaze are consistent with Lévy superdiffusion. Finally, Section 3.4 presents the spatiotemporal predictions of the model across these scales and discusses how they unify the experimental and modelling findings above.

3.1 Theta rhythmicity of saccades

The approximate 200 ms timescale of human saccades has long existed experimentally [25, 168–170]. A direct measure of saccade rhythmicity is the spread of the ISI distribution, assuming a large log-normal spread of intervals Δt. This can be used to construct a measure called the rhythmicity index (RI):

$$\mathrm{RI} = \frac{1}{\mathrm{CV}_{\ln}}, \qquad \mathrm{CV}_{\ln} = \sqrt{e^{\sigma_{\ln}^2} - 1} \tag{3.1}$$

where $\sigma_{\ln}$ is the standard deviation of the distribution of ISIs in log-space $\ln \Delta t$, with $\mathrm{CV}_{\ln}$ often called the geometric CV. This RI is an effective measure of the spread of a strictly positive distribution. A Poisson distributed variable will give $\mathrm{RI} = \frac{\sqrt{6}}{\pi} \approx 0.7800$[1], whereas it is found that for the ISI distribution of subjects considered here for a random NOISE condition is $\mathrm{RI} = 2.69$ [61].

Strong rhythmicity of saccades with a slowly changing theta rhythm

The recent discovery of the strength of saccade rhythmicity by Näher et al. [61] and others [171] required a deep understanding of the rhythmicity of biological systems. A challenge that is often present in biological systems is that timescales are not driven by a perfect clock, but instead the characteristic timescales of certain regular patterns in the brain. As such, the 200 ms estimate for saccades during a visual search is not expected to be a universal timescale across subjects and trials. Instead, the model used for such processes in neuroscientific works is a noisy rise-to-threshold model, where a random walk is constructed, and when it reaches a goal, a decision is made [61] [CITE]. The noise-driven rise for this process is expected to have a rate which is slowly varying about a characteristic timescale, in the case of saccades around 2–6 Hz. Since this slowly varying rate is expected to be stable for a single individual and a single trial, we consider a local measure of rhythmicity for the intervals between the saccades within a timseries, called the Inter-Saccadic-Intervals (ISI). Such a metric can

[1] This can be found from the exponentially distributed intervals of a Poisson process with mean λ as $X \sim \lambda e^{-\lambda X}$, changing variables to find the distribution of $\ln X = Y \sim \lambda \exp[Y - \lambda \exp Y]$. This distribution will have $\sigma_{\ln} = 0.986$

be proposed through the local Rate-Corrected Rhythmicity Index (RCRI) [61]:

$$\mathrm{RCRI} = \frac{1}{1 + \langle \mathrm{CV}_2 \rangle}, \qquad \mathrm{CV}_2(i) = 2\frac{\|\Delta t_{i+1} - \Delta t_i\|}{\Delta t_{i+1} + \Delta t_i} \tag{3.2}$$

which allows quantification of purely local rhythmicity, comparing each interval Δt_{i+1} to it's predecessor Δt_i, and averaging across all intervals considered i, usually a single subject. The results of such an analysis have resulted in two key conclusions. Firstly, human saccades are very strongly rhythmic, with RCRI ≈ 0.76 for the subjects as described in Section 2.1 depending on condition. Secondly, it has been found that the instantaneous saccade rate is a strong predictor for the local saccade rhythmicity RCRI, with a correlation calculated at $\rho = 0.39, P = 0.001$, with saccade rate found to be a significant predictor of RCRI with $\beta = 0.0114\,\mathrm{Hz}^{-1}, P = 0.014$ [61]. It has been found that those trials binned into the low-, medium-, and high- saccade rate tertiles will show a positive correlation of saccade rate with higher RCRI [61], as shown in Section 4.1.1. A table comparing these indices and rhythmic patterns is also shown below in Table 3.1

Table 3.1: Effect of saccade rate on rhythmicity and ISI peak.

Saccade Rate	RCRI	RI	⟨ISI⟩	Saccade Rate	RCRI	RI	⟨ISI⟩
$\mathbf{t}_\theta = 213.33\,\mathrm{ms}$	0.6787 [0.675–0.683]	2.364 [2.33–2.40]	195.0 ms	Low	0.71	2.76	210 ms
$\mathbf{t}_\theta = 177.78\,\mathrm{ms}$	0.6857 [0.685–0.692]	2.237 [2.20–2.28]	178.4 ms	Medium	0.79	3.43	235 ms
$\mathbf{t}_\theta = 142.22\,\mathrm{ms}$	0.6990 [0.696–0.702]	2.116 [2.08–2.15]	152.7 ms	High	0.85	3.60	260 ms

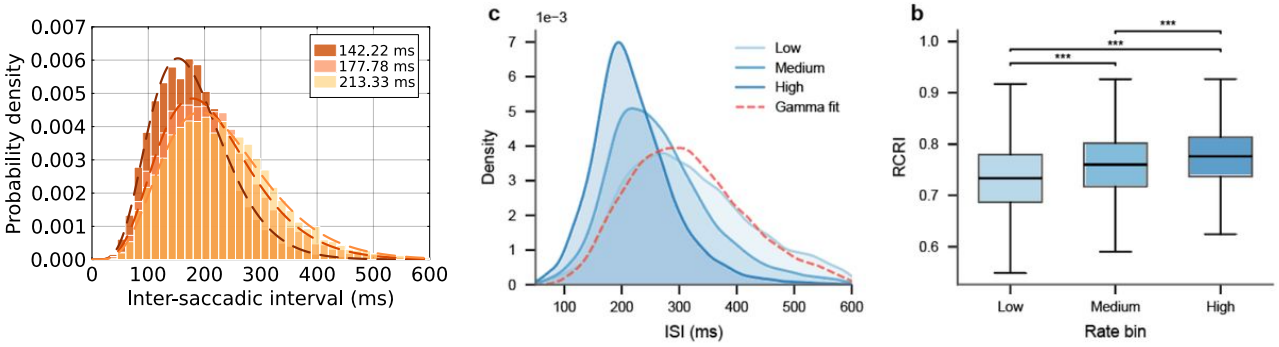


Figure 3.1: **(a)** shows the distribution of ISIs given by the presented model, and shows the Gamma-renewal model fit. For the given parameters (as in Section 2.1), the fit gives RCRI $= 0.695, k = 6.479, \langle \mathrm{ISI} \rangle = 148.5\,\mathrm{ms}$. This is compared to the theta rhythm given by our model parameters of $t_\theta = \tau \times \sigma_\mathrm{k}^2/\sigma_\mathrm{fix}^2 = 142.2\,\mathrm{ms}$. We thus find that the model produces a high degree of rhythmicity from the adaptive kernel added to the model. **(b)** shows the result that trials binned with shorter ISIs are correlated with a high rhythmicity of saccades ($\rho = 0.39, P = 0.001$). Such a conclusion corresponds to the kernel dynamics presented, where faster kernel dynamics will increase rhythmic transitions (see Section 2.2.3). **(c)** presents the empirical distribution of ISIs after trials are binned into low- medium- and high-tertiles. ***(b)*** *and* ***(c)*** *adapted from Figure. 5 of Näher et al. [61]*

Interpretations of local rhythmicity and saccade rate for a model of gaze dynamics

These results support our unified stochastic-informative formulation of gaze dynamics. We have shown good agreement between the tight distribution of modelled ISIs, more closely locally. We have also introduced a natural parameter τ with which to modulate the slowly varying theta-scale saccade rate. This also connects the dynamics of information gathering with the dynamics of regular attention in a search task. The results that rhythmicity is strongest where a task contains nearly identical searched objects is encouraging for this information-driven saccade rate theory. A correlation between high saccade rate and high rhythmicity is predicted from the origin of regular dynamical-informational

transitions as described in Section 2.2, where a stronger kernel produces more rhythmic transitions, and a weaker kernel will approach an arrhythmic Poisson point distribution of ISIs, reducing the RCRI, as predicted.

3.2 Gamma oscillations during visual fixation

We now look to verifying the short-time and fine spatial scales of our model predictions, specifically of those high frequencies found in local sampling of potential modes in Section 2.3. Typically short-time gaze dynamics are divided into microsaccades, low-frequency linear drift, and high frequency ocular tremor [41]. There is increasing evidence that microsaccades—which is the name given to jumps in gaze during fixaton—are driven by the same statistics and process as regular (called macro-) saccades. This is predicted by our model, where saccades on all scales are manifestations of the Lévy flight jump process; a full analysis of the superdiffusion of such saccades on the micro- and macro-scales is covered in the next section (Section 3.3). Linear drifts in fixation have been found to be important in gathering information. An account of low-frequency linear drift easily fits into the model presented here, for instance as steep peak in the dynamical kernel or a simple drift in the potential landscape, producing a self-avoiding walk as described in [7] and described in Table A.1. But our focus here is that of the high frequency signal called microtremor found in eye movements [67, 172, 173] and misattributed to noise [67, 72, 174], but which we predict to be important to the function of a visual search.

The current study is the first to make a well-motivated prediction concerning high-frequency ocular microtremor as a gamma-scale sampling pattern linked to deeper neural activity. Identifying ocular tremor as high frequency gamma-scale visual sampling is a novel and useful approach in understanding and quantifying perceptual sampling. This is supported by recent evidence that ocular microtremor may have a shared origin for both eyes in the brainstem [67], and coarse results that neural disorders such as Parkinson's' disease effect tremor frequency across all tasks [175].

3.2.1 Previous treatment as ocular microtremor

Ocular microtremor is the smallest-amplitude component of fixational eye motion. It has been observed across experimental studies, with reported amplitudes of up to 30–40 sec of arc [58, 176]. Despite its small amplitude, microtremor has been proposed to contribute to the dynamic sampling of high-contrast features and fine spatial detail [27, 70–72]. However, a recent physical study concluded that its magnitude may be insufficient to produce a meaningful change in the retinal image during prolonged fixation [177]. We therefore consider the hypothesis that the role of microtremor will be task-dependant: it may have a more important functional role during visual search, when gaze is rapidly adjusted in response to incoming information, than during prolonged, stationary fixation. Further, the role of microtremor has been demonstrated to be important in accounting for further brain function, such as being a biomarker for neurological fatigue [178], brain state during anaesthesia (under which the tremor frequency will halve) [179–181], and as a biomarker for schizophrenia [182].

In our search-task data, the tremor spectrum has a broad peak at a lower frequency than is typically reported, where we find a peak at 56 Hz, and with a larger estimated amplitude. By comparison, studies of longer fixation periods generally report more localised peaks around 80–90 Hz [67].

The present analysis differs from the existing literature in two important respects:

1. We examine eye movements during a naturalistic visual search task. The major studies in this field, as for instance surveyed in the influential review by Graham et al. [67], instead measure tremor across a prolonged fixation, typically lasting more than 3000 ms [57, 58, 176].

2. The data were not collected specifically to measure this high-frequency component. Ocular microtremor is commonly studied using velocity or acceleration signals, which emphasize high-frequency motion. In our data, the estimated tremor amplitude is larger than values often reported: approximately 200 sec of arc, compared with values of up to 40 sec of arc in prior

studies [58, 176]. Because the recordings and analyses were not optimised for direct comparison of tremor amplitude, this difference should be interpreted cautiously.

The fact that the following analysis can be done on the positional data motivates a closer study on the fixational tremor frequency during the rapid fixations which occur during visual search. In particular, testing the hypothesis that this high amplitude gamma-scale microtremor is a sampling process can be tested rigorously through experimental designs presented in Section 3.4.

3.2.2 Analysis of fixational power spectra

We analyse the spectral properties of fixational gaze trajectories during visual search. The shorter, more naturalistic fixations in our data pose a challenge for estimating high-frequency motion. In our data collected from Näher et al. [61] fixations contain on average approximately 200 ms of data, which yields only about ten cycles of a 50 Hz signal. Moreover, estimating high-frequency motion from noisy position measurements can be dangerous, as the white noise often present in these detectors often approaches tremor amplitude [57, 58].

Our recordings were sampled at 500 Hz, with reported positional precision of up to 36 sec of arc [183]. Because low-frequency drift can dominate the positional power spectrum, we analyse velocity rather than position. The expectation of the position spectrum is that it will scale approximately as $\mathrm{PSD}_{\mathbf{x}}(f) \sim f^{-2}$, which is the assumption for tremor-less gaze fixations modelled with Brownian noise [7, 69]. Velocity, as found through differentiation will multiply spectral power by a factor by $(2\pi f)^2$, yielding an approximately flat velocity spectrum with no peak. Thus, under a baseline model with no distinct tremor component, we would expect no peak in the velocity spectrum, instead just a flat baseline with $\mathrm{PSD}_{\mathbf{v}}(f) \sim f^{0}$.

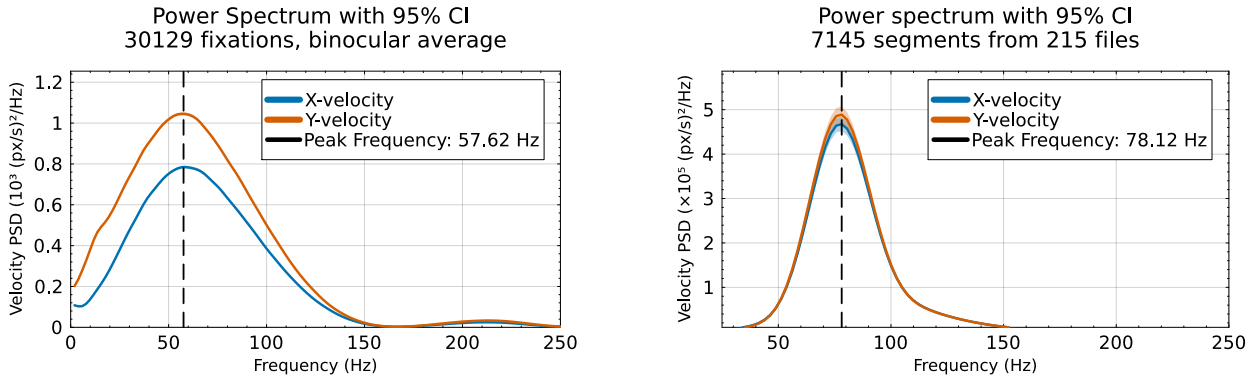


Figure 3.2: Comparison of power spectra from fixations in eye-tracking data and modelled trajectories. a) The power spectrum taken from the fixational periods of human gaze trajectory data provided by [61]. A clear broadband peak is found with a peak centre at $f = 57.62$ Hz. **b)** The power spectra of fixations as applied to the fixations in the modelled gaze trajectories. A similar broadband peak is found, with peak frequency $f = 78.12$ Hz. *Methods used to pre-process data and construct these power spectra are to be found in Section 3.2.2.*

Data processing

The analysis pipeline is as follows, and is identical for both the human eye-tracking data and the model gaze trajectory unless otherwise stated. We collect all fixations across the datasets, either as they are previously labelled in the eye-tracking data, or else by using a variation of the median-filtering method presented in the literature [184, 185], which was verified manually and can be found implemented in `VisualFNS_Analysis.jl` (Chapter B). After truncating at any NaN valued, any fixations shorter than 100 ms were not considered. We apply a fourth-order Butterworth low-pass filter with a cutoff of 200 Hz to reduce high-frequency instrumental noise near the Nyquist frequency $f_{\mathrm{Nyquist}} = 250$ Hz in

both datasets, as high frequencies will amplify after the following numerical differentiation. We then estimate the two-dimensional velocity by central differencing the filtered position data, and average between the two eyes for the eye-tracking data. Welch's method was then used to construct the power spectra. A Hanning window was used of length 256 ms (256 samples) was used, with an overlap of 64 ms (64 samples). These spectra are shown in Figure 3.2 with their ensemble size and confidence ribbons.

Importance and discovery of gamma-scale fixational dynamics

The existence of a high frequency ocular microtremor has been documented in detailed analyses of visual fixation [39, 67, 72], and in studies dedicated to their measurement [172, 175]. However, this is the first study to provide a mechanistic prediction of tremor as coherent gamma-frequency exploration of a visual scene. Moreover, this study is the first to study ocular microtremor in the positional measurement of the eye during a search task. For the interested reader, a typical fixation trajectory and its Fourier decomposition are provided in Figure C.1. The major difference in the current results compared to other studies of OMT are that the peak frequency is significantly smaller than most estimates, which generally place mean tremor frequency in the range 80–90 Hz, as well as being slightly lower power. A lower frequency here may suggest a different neurological dynamics during visual exploration compared to a prolonged fixation. This can be compared to the result that the gamma frequency in the brain is found to be higher when viewing local areas of high-contrast [26, 186, 187]. A physically motivated explanation here may be that the tightly localised attentive potential wells during fixation produce attenuated high frequency signals to ocular muscles, whereas during a search task the unexplored and dynamically salient area promotes higher amplitude low frequency OMT. For a further discussion of the predictive power of this interpretation of OMT as gamma-scale sampling, and experiments and studies which are designed to test this theory, see Section 3.4.

3.3 Search superdiffusion

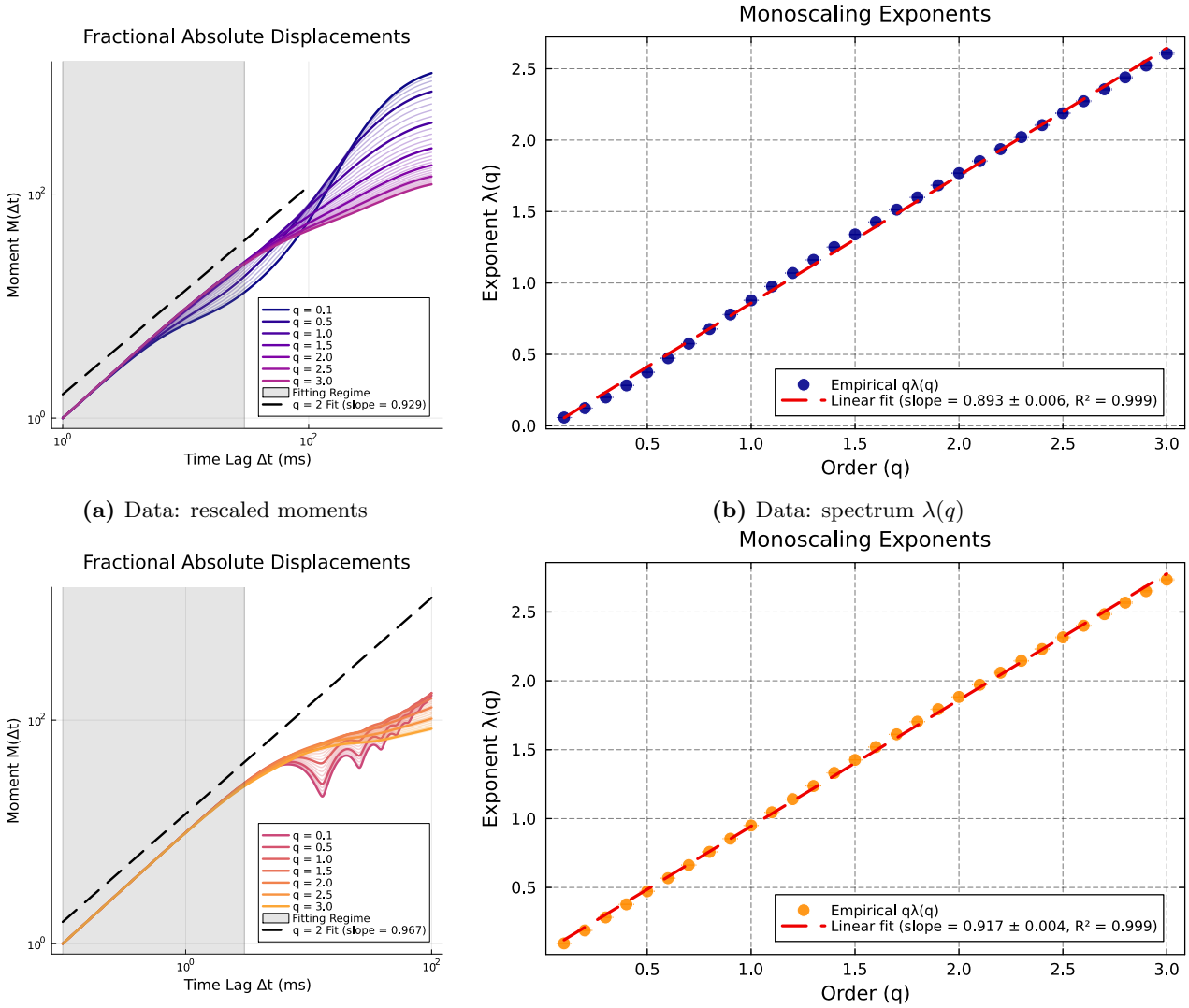
Superdiffusion is often a prominent pattern in natural dynamical systems, where its existence implies some self-similar fractal structure. Such dynamics have not been the focus of other studies, and often visual trajectories have been considered as a fixational Brownian motion externally interrupted with saccades. We here present novel multiscale superdiffusive pattern in the diffusion of eye movements from eye trajectory data. Many studies which have described eye movement dynamics as normally diffusive stems from the sensitivity of data processing choices in finding anomalous diffusion exploring confined boundaries such as fixations. Conventional analyses often inadvertently obscure critical spatiotemporal features.

The most important methodological choice in accurately characterising visual diffusion patterns is the choice to time-average displacements, and choosing a short timescale over which diffusion scaling is considered. As demonstrated by Vilk et al. [115], aggregating data across disparate temporal regimes can wash out localised scaling behaviours, yielding an artefact of normal diffusion. Studies which construct regimes of normal and subdiffusive regimes are thus common when examining eye movements, but this ignores the continuous-time superdiffusion which we unify under multiple scales. For such studies, see [22, 44, 72, 188].

Analysis of multiscale superdiffusion in eye-tracking data

We analyse the binocular-averaged gaze trajectories over all 22 human subjects provided from the study in Näher et al. [61], as described in Section 2.1. From the onset of the search task, we time-average the fractional absolute displacements $\langle \|\Delta \mathbf{x}(\Delta t)\|^q \rangle$ for 30 values of q linearly separated from $q = 0.1$ to $q = 3.0$. These fractional absolute displacements are evaluated across all valid time lags $\Delta t < 1000$ ms with a resolution of $\Delta t = 1$ ms. All intervals containing NaN values (such as blinks) are discarded. We then ensemble-average these displacements across all subjects.

We perform a linear best-fit in log-log space of these absolute fractional displacements to find power-law diffusive scaling with time, thus sampling from our moment spectrum function $\lambda(q)$ as defined



(a) Data: rescaled moments

(b) Data: spectrum $\lambda(q)$

(c) Model: rescaled moments

(d) Model: spectrum $\lambda(q)$

Figure 3.3: Monoscaling superdiffusion in the model predict a single superdiffusive exponent across scales in human visual search trajectories. Time-averaged fractional absolute displacements $\langle \|\Delta\mathbf{x}\|^q \rangle$, which generalise the mean squared displacement to a spectrum of orders q, with small q emphasising small fixational displacements and large q emphasising saccades. The top row shows binocular-averaged gaze data, with moments ensemble-averaged over all 22 subjects of Näher et al. [61] (approximately 10,000 s of data; exact sample size to be updated). The bottom row shows the same analysis applied to trajectories simulated from the model of Chapter 2. **(a,c)** Moments rescaled as $M_q(\Delta t) \propto \langle \|\Delta\mathbf{x}\|^q \rangle^{1/q}$ and normalised to $M_q(1\text{ ms}) = 1$, for 30 values of q from 0.1 to 3.0. In log–log space the slope of each curve is $\lambda(q)/q$. **(b,d)** Scaling exponents $\lambda(q)$ from power-law fits to the curves on the left, up to $\Delta t = 30\,\text{ms}$. For the data, the relation is linear, $\lambda(q) = (0.893 \pm 0.006)\,q$ with $R^2 = 0.999$, so all moments share a single scaling exponent. This corresponds to a Lévy stability index $\alpha = q/\lambda(q) = 1.120 \pm 0.007$, well below the Brownian value $\alpha = 2$, indicating strong superdiffusion from fixational scales to saccades. For the model, the fit gives $\lambda(q) = (\,0.917 \pm 0.004\,)\,q$, i.e. $\alpha = 1.091 \pm 0.005$.

in Equation (1.6). Each of these fractional absolute displacements are shown in **??**. The moment spectrum function $\lambda(q)$ thus is found to have a good fit to $\lambda(q) = (0.893 \pm 0.006)q$ with an error in the fit of $R^2 = 0.999$. This behaviour gives monoscaling across the shortest to longest spatial scales of fixations to saccades. The corresponding Lévy exponent given such linear monoscaling in $\lambda(q)$ is thus found to be $\alpha = \frac{q}{\lambda(q)} = 1.120 \pm 0.007$.

3.3.1 Spectrum of fractional moments

Figure 3.3 summarises the scaling of the time-averaged fractional absolute displacements. For comparison of the various scaling exponents, we define a rescaled moment function $M_q(\Delta t)$ to give good comparison of the spectrum of moments. $M_q(\Delta t)$ is here defined as

$$M_q(\Delta t) = M_q(\Delta t_{\min})^{-1} \langle \|\Delta \mathbf{x}(\Delta t)\|^q \rangle^{1/q} \sim \Delta t^{\lambda(q)/q} \tag{3.3}$$

with C_q defined such that $M_q(\Delta t_{\min}) = 1$ to give a shared origin at the first time lag Δt. Small q gives strong weight to the small, frequent displacements of fixational motion, while large q weights the rare, large displacements which are given in saccades. The shape of $\lambda(q)$ therefore tests whether one exponent may describe superdiffusion on all scales. A linear spectrum, $\lambda(q) = q/\alpha$, is monoscaling: all moments grow with the same exponent, as for a self-similar Lévy process with stability index α. A nonlinear spectrum is the general case found in many systems called strong anomalous diffusion which consist of two dynamical strategies stitched together at different scales [115, 166, 189–191]

Figure 3.3a shows the rescaled moments $M_q(\Delta t)$ for the human eye-tracking data. Under this rescaling, the slope of each curve in log–log space is $\lambda(q)/q$. Monoscaling therefore predicts that these curves share a single slope across the whole spectrum of q, with $\frac{\lambda(q)}{q} = 1/\alpha$. For comparison, Brownian motion ($\alpha = 2$) has slope $1/2$ and ballistic motion has slope 1 across the entire spectrum q. The monoscaling exponent is found by performing a linear regression to our spectrum $\lambda(q)$. Figure 3.3b shows these fitted exponents $\lambda(q)$ against q. The linear fit is good, with $\lambda(q) = (0.893 \pm 0.006)\, q$ and $R^2 = 0.999$, across the whole considered range $0.1 \leq q \leq 3.0$. This gives a Lévy index $\alpha = q/\lambda(q) = 1.120 \pm 0.008$, which is well below the Brownian value of 2 and so indicates strong superdiffusion. Because the spectrum is well described by a linear regression, there is no crossover in scaling between the low-q moments that probe fixational motion and the high-q moments that probe saccades. A single exponent describes both, validating our unifying model across both scales of study in gaze dynamics, derived naturally from FNS dynamics.

Implications of monoscaling superdiffusion

The most important implication of this particular superdiffusion results is that the strong monoscaling of visual search trajectories with Lévy index $\alpha = 1.120$ is robustly multiscale. The robust linear scaling of $\lambda(q) = \frac{q}{\alpha} = \frac{q}{1.120}$ across a spectrum of q-values $0 < q < 3$ is important, as it presents an identical process driving both fixational eye movements — including tremor, drifts, and microsaccades — with small relative values of $\|\Delta \mathbf{x}\|$ emphasised with low values of q, as well as large nonlocal saccades with large relative values of $\|\Delta \mathbf{x}\|$, emphasised through large values of q. Importantly, this validates our construction of a continuous-time unified model of eye movements, unified across all spatiotemporal scales of visual search.

3.4 Spatiotemporal predictions from the model

Throughout this chapter, we present a novel physics-inspired analysis of a standard gaze-tracking experiment during visual search which yields new results and predictions of gaze dynamics. In particular, the approximate timescale of saccades has been known since the beginning of the field of eye movement research, whereas the significance of the rate-corrected local rhythmicity is a very recent result only discovered in the year of this present work. OMT as a phenomenon has been found in the literature [67], however no consensus exists as to it's importance in the functioning of gaze, and it has never been linked to a computational role of gamma-scale rhythms. Lastly, the mathematical analysis

of search superdiffusion of gaze has only been recently undertaken in the literature, and without the well-motivated link to a model of gaze we present here. However, another benefit of comparing the generative unifying model presented in Chapter 2 to the data is that further dynamical predictions outside of the present literature can be made. Thus, we present experimental paradigms and features of this model which are being developed to better understand visual searching.

3.4.1 Saliency-frequency coupling in contrast-dependant search tasks

OMT frequency is here hypothesised to be the result of a harmonic oscillations in the control of the path of the eye during fixation tasks. An experiment may be designed in which targets with various degrees of contrast are presented to be searched. High contrast, localised points, such as tight, high spatial frequency Gabor patches, would produce localised saliency wells with low σ_{fix}. Whereas, low contrast, large low spatial frequency patches would produce saliency wells with a large width σ_{fix}. High contrast wells would correspond to shorter theta timescales, quadratic in the effective saliency well size as in Equation (2.8), and a microtremor rhythm which is approximately inverse to the effective saliency well size as in Equation (2.11). The effect of visual contrast has been found to be significant in gamma frequency in the visual system, and this quantitative relationship provides an experimental prediction of the relationship between three data points: t_θ, f_{peak}, and k_{Gabor}. A simple regression analysis can be achieved here to find correlations between spatial frequency k_{Gabor} and the visual theta-rhythm t_θ and the f_{peak}. It can then also be tested if these effects are consistent with a potential well of variable size, such that $t_\theta \propto \sigma_{\text{fix}}^{-2}$ and $f \propto \sigma_{\text{fix}}^{-1}$

3.4.2 Smooth pursuit predictions and parametric-fitting techniques

Another experimental paradigm which provides insight into the behaviour of the control of visual trajectories. In particular, the exact same model can be examined in the behavioural paradigm of smooth tracking of a moving object. In particular, a typical experiment would be the tracking of a smoothly moving pendulum. It is realistically found that a typical eye movement trace contains discontinuous micro-saccades during smooth pursuit, and discontinuous catch-up saccades. In particular, we expect from the model that the instability will be most present in the periods of stability when kernel deposition is strongest. Further experiments could also find the behaviour of dynamics in phase space $\alpha/\beta/\gamma/t_\theta$ for different neurological conditions. For instance, it is well-known that schizophrenia effects the

Summary

After examining our proposed model in depth, we have now compared to real human gaze data. This work has provided a dynamical explanation of the recently discovered significant regularity of saccades during a search task as a coupling between model dynamics and information through kernel deposition. We then made the novel prediction of gamma-scale oscillations in fixation dynamics, which was validated under the present data analysis. This component of fixational dynamics has rarely been studied as a coherent aspect of neural dynamics; we are the first to provide a modelled explanation of its neural origin. Finally, we presented the prediction of our model of unified fluctuation-driven superdiffusive motion across all spatial scales. By analysing fractional q–moments, we found that gaze trajectories are characterised by a single superdiffusive exponent across all scales, unifying the dynamics of fixations and of saccades under a single model. Finally, we suggest further experimental methods to test further quantitative predictions of the model designed to examine attention in the brain through the study of gaze dynamics.

Chapter 4: Functional Advantages of Visual Search Dynamics

Now that we have developed and validated a rigorous model of the dynamics of gaze across all spatiotemporal scales, we investigate what we can learn from the efficiency of the model to an informative search task. The first section of this chapter (Section 4.1) presents informational-theoretic methods which have been constructed to analyse efficiency during abstract search tasks. Key to this method is the quantification of information of a belief distribution as the distribution entropy $S_{\pi_t} = -\int_{\mathbf{x}} d\mathbf{x} \ln \pi_t(\mathbf{x})$, and tracking its development through a search task. We take these metrics and compare the informational advantages of each dynamical pattern by comparing an abstract sampler operating in different points in parameter space $\alpha/\beta/\gamma/\tau$. The final section of the main body of this thesis (Section 4.2) examines further extensions of these dynamics beyond the case of a 2-dimensional visual search to high-dimensional or dynamical target landscapes, where control and exploration are hypothesised be more responsive and robust to shifts in a loss function.

4.1 Functional roles of dynamical sampling patterns

The dynamics studied in this thesis are characterised by three distinct spatiotemporal signatures: theta-scale rhythmicity, gamma-scale local oscillations, and superdiffusive spatial trajectories. The framework of Chapter 2, constructs a biologically-motivated model to produce these signatures through their emergent physical dynamics. Throughout Chapter 3 we have found that the same dynamics have been found to be important in studying real human gaze trajectories, especially during realistic search tasks. In this section we make use of the theoretical foundations of this model as a sampling theory, and identify the functional advantages of each pattern in an abstract sampling process within neural computation. To quantify the role each signature provides, we define a set of information-theoretic metrics for search efficiency and, for each metric in turn, describe how each of the three patterns effects it.

Information gain

When considering the efficiency advantages of strategic sampling in a visual search task, the central quantity under study is the information gain, $\mathrm{IG}(t)$. To quantify informational across an abstract search landscape, we take inspiration from Bayesian inference and sampling methods. In particular, the link between the explored dynamic potential to a target-belief distribution in a search task in Equation (1.3) provides a natural place to define information gained in the target search.

A natural first step in comparing information gathered about a target position through time across a target is the cross-entropy of the posterior to the prior,

$$H(\pi_t, \pi_0) = -\int \pi_t(\mathbf{x}) \log \pi_0(\mathbf{x})\, d\mathbf{x}. \tag{4.1}$$

however, this cross-entropy alone is of limited utility. Cross entropy will conflate the internal concentration of belief within the posterior (its self-entropy, $H(\pi_t)$) with the net information gained relative to the prior. To decouple these contributions we subtract the posterior entropy to obtain the Kullback–Leibler (KL) divergence $D_{\mathrm{KL}}(\pi_t \parallel \pi_0)$, which gives the formal definition of information gain

up to time t:

$$\begin{aligned} IG(t) &= H(\pi_t, \pi_0) - H(\pi_t) \\ &= -\int \pi_t(\mathbf{x}) \log \pi_0(\mathbf{x})\, d\mathbf{x} + \int \pi_t(\mathbf{x}) \log \pi_t(\mathbf{x})\, d\mathbf{x} \\ &= \int \pi_t(\mathbf{x}) \log \left(\frac{\pi_t(\mathbf{x})}{\pi_0(\mathbf{x})} \right) d\mathbf{x} \\ &\equiv D_{\mathrm{KL}}(\pi_t \parallel \pi_0). \end{aligned} \tag{4.2}$$

where important properties of the KL divergence is that it is non-negative and asymmetric. In the context of an abstract sampling model, non-negativity reflects the fact that any spatial update from π_0 represents accumulated evidence, though not necessarily optimal search efficiency. A weak kernel, a slow informative timescale τ, or deposition in low-likelihood areas alters the landscape only marginally and results in slower growth in IG. The information content of the search trajectory is therefore entirely captured by IG, which quantifies the global information acquired as the observer's belief landscape evolves from π_0 to π_t. The metrics in the following subsections summarise different aspects of the curve $\mathrm{IG}(t)$.

4.1.1 Results

Metrics and comparison conditions

We collected statistics on an ensemble of searches on the computational setup described in Section 2.1, with 200 trials for each parameter set examined, across the entire swept range of parameters. We considered two sweeps: $\alpha - \beta$ sweep for α incremented from 1.1 to 2.0 in increments of 0.05 and β from 0 to 1200 in increments of 100, and a $\alpha - \theta$ sweep with the same α range, and θ varied between 5×10^{-5} and 1×10^{-3} in increments of 5×10^{-5}

The most important three metrics that capture complementary aspects of performance: the speed of completion as first passage time, the speed of information gathered as information velocity, and the temporal profile of information gain as a shape exponent. The first passage time (FPT) is the time from trial onset to the first fixation on a target, perhaps the clearest metric of success, with units of seconds. The information velocity $\mathrm{IV} = \langle \mathrm{dIG}/\mathrm{d}t \rangle$ is the mean value of the rate of information gain, with units of inverse seconds. The shape exponent SE describes the profile of the IG curve: we normalised time and IG to $[0, 1]$, computed the area A under the normalised curve, and defined $\mathrm{SE} = 1/A - 1$. This will fit the monotonic information curve the exponent of the power law $\mathrm{IG} \propto t^{\mathrm{SE}}$ with equal area. Thus $\mathrm{SE} = 1$ is accumulation at a constant rate, $\mathrm{SE} < 1$ is decelerating information gain, and $\mathrm{SE} > 1$ would be an accelerating rate of information gain.

As an analytical reference for FPT, we considered a searcher that fixates points uniformly at random without replacement, one per theta interval. For K targets among N points, the expected number of fixations to the first target is $(N+1)/(K+1)$[1], and so for $\tau = 2 \times 10^{-4}$

$$\mathbb{E}[\mathrm{FPT}_0] = t_\theta \frac{N+1}{K+1} = 6.0919\,\mathrm{s} \tag{4.3}$$

where t_θ is the mean saccade interval $t_\theta = 142.22\,\mathrm{ms}$.

Summary of parameter effects

Table 4.1 summarises the change in each metric across the tested range of each parameter, relative to baseline.

[1] This is found by considering the expected number of failures in sampling from a hypergeometric distribution with total population N and success population K. See for instance [192]

Table 4.1: Correlations between key dynamical parameters and information metrics. Correlations are computed between each parameter and the median or mean of each metric across $n = 250$ trials. 95% bootstrap confidence intervals are shown in brackets below each estimate.

	Median			Mean		
Parameter	ρ_{FPT}	ρ_{IV}	ρ_{SE}	ρ_{FPT}	ρ_{IV}	ρ_{SE}
Local oscillations β	-0.300	0.641	0.508	-0.475	0.767	0.812
	$[-0.633, 0.091]$	$[0.287, 0.891]$	$[0.148, 0.797]$	$[-0.720, -0.176]$	$[0.445, 0.951]$	$[0.577, 0.945]$
Kernel rhythmicity	—	—	x	—	—	x
			$[\cdot, \cdot]$			$[\cdot, \cdot]$
Superdiffusion α	0.554	-0.748	-0.761	0.790	-0.951	-0.880
	$[0.262, 0.712]$	$[-0.898, -0.544]$	$[-0.895, -0.607]$	$[0.640, 0.898]$	$[-0.981, -0.911]$	$[-0.953, -0.775]$

Theta rhythmicity (τ)

The strength of rhythmicity between points of interest is difficult to quantify, since any modulation of kernel dynamics will trivially change the dynamics of information gathering, as our model couples rhythmicity with information gathering. Thus, we examine rhythmicity only through a single metric which has been decoupled with time: the shape exponent SE.

Local oscillations (β)

Local momentum-driven oscillations have here been shown to both improve local sampling efficiency, as well as global responsiveness to transitions. Local sampling can be measured by the information velocity. Our results show that IV has the highest correlation with β, supporting this hypothesis. However, global search efficiency benefits from the momentum-driven dynamics, as shown in the positive correlations with FPT and SE.

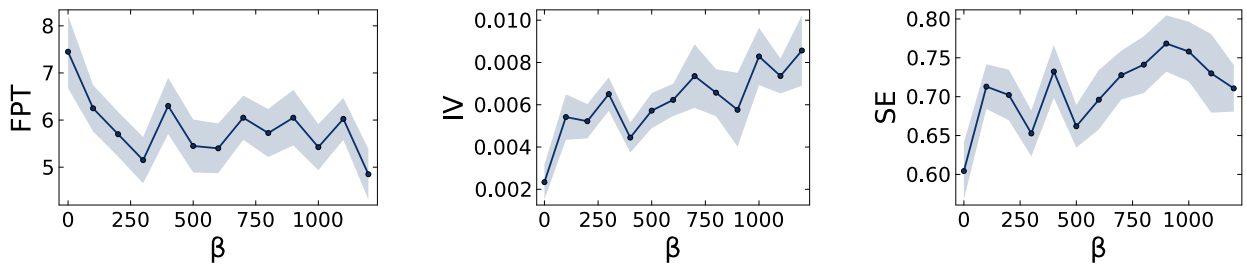


Figure 4.1: Trends in three key information metrics with local oscillations.

Superdiffusion (α)

We recover the result that a sparse search for a target is optimally performed under a superdiffusive trajectory, even combined with our additional modelled dynamics. This is robustly seen in the strong positive correlation between α and FPT, as well as the strong negative correlations between α with both IV and SE.

Summary

4.2 Further extensions of studied dynamics

The three patterns studied in this thesis were introduced to explain gaze, but each is a general mechanism for exploration of a sparse and unknown landscape: nonlocal superdiffusive jumps, nested oscillations that organise local sampling and transitions between regions of interest, and a history-dependent kernel that records information gathered. Nothing in their construction is specific to vision,

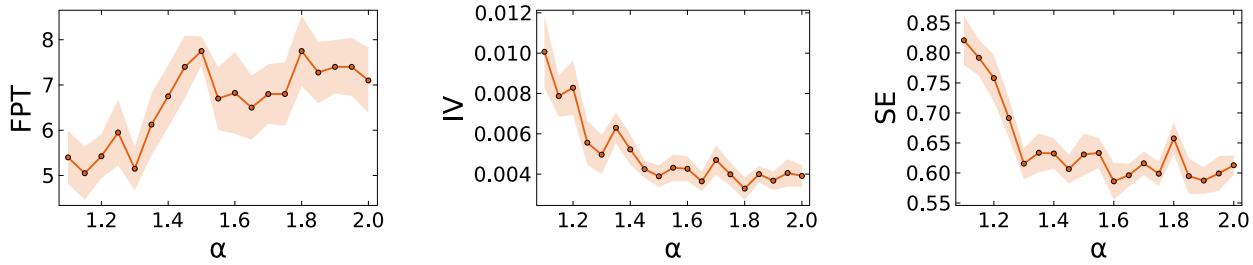


Figure 4.2: Trends in three key information metrics with global superdiffusion.

and the model we have studied can be extended to a parameter space of any dimension. In this section we outline how they might carry over to two settings in machine learning and artificial intelligence, namely sampling from difficult probability distributions and pursuing goals that change in time. These are proposals for future work and are not tested here.

4.2.1 Superdiffusive probabilistic sampling

A central task in Bayesian inference and generative modelling is to draw samples from a distribution $\pi(\mathbf{x}) \propto \exp\{-V(\mathbf{x})\}$ that may be multimodal, with well-separated modes, or heavy-tailed. Standard gradient-based Markov chain Monte Carlo (MCMC) methods, such as overdamped Langevin dynamics, are driven by Brownian noise and so explore locally. The time to leave a mode grows exponentially with the height of the barrier around it (Kramers' law), so samplers can remain trapped in one mode for very long periods, and the weak gradients in heavy tails make the dynamics inefficient there too [[**CITE**]]. The same problem is solved by the gaze model: a searcher must visit sparse regions of high likelihood across a landscape without being trapped by any of them.

Each pattern studied here addresses part of this problem. First, driving the dynamics with α-stable Lévy noise ($\alpha < 2$) replaces local diffusion by a power-law distribution of jump lengths. For Lévy flights the escape from a well is governed mainly by the distance to the next mode, rather than by the height of the barrier between them [[**CITE**]], so jumps between separated modes occur at a rate that does not decay exponentially with the barrier. The theory of fractional neural sampling (FNS) was formulated for exactly this purpose, and was shown to sample efficiently from multimodal distributions [33]. Second, the memory–momentum term produces fast local oscillations within a mode, which decorrelate nearby samples and probe the local gradient of π. Third, the dynamical-informative kernel of Section 2.2 adds to the potential

$$V_t(\mathbf{x}) = V_0(\mathbf{x}) + \int_0^t k(\mathbf{x}_{t'}) \, \frac{dt'}{\tau}, \tag{4.4}$$

which penalises revisited regions. This closely resembles well-tempered metadynamics [193], in which deposited bias fills explored minima and pushes the sampler towards new ones. The theta-rhythmic transitions provide a fourth ingredient: sampling alternates between phases of local exploitation and phases of nonlocal movement within a single process, rather than through a hand-designed mixture of local and global proposals.

4.2.2 Dynamical goals

In the search tasks studied here the goal is fixed: the likelihood landscape $V_0(\mathbf{x})$ is set before the search begins, and only the observer's belief about it changes. Many problems in machine learning and artificial intelligence instead have goals that vary in time. Targets move or appear, rewards shift, and an agent switches between tasks. Handling these requires a search that can both remember what it has explored and abandon that memory when the objective has changed. This is the situation of the visual system in the active-vision loop, in which the observer's goals and the scene change during a search, and it suggests a natural extension of the model to time-dependent landscapes $V_0(\mathbf{x}, t)$.

The kernel provides a mechanism for this. Currently deposited information accumulates for the whole trial, so a revisited region stays penalised. Adding a finite memory, in which the deposited kernel decays at a rate γ, creates a trade-off controlled by one parameter. Slow decay strongly avoids revisiting, which suits a static goal, while fast decay allows the sampler to return to regions that may have changed, which suits a moving or changing goal. The other patterns would then have clear roles. The theta rhythm gives a clock on which the current goal is re-evaluated, and the local oscillations measure its gradient, in analogy with the tracking of a moving target in smooth pursuit. Superdiffusive jumps provide a way to reacquire a goal after it has moved far away, a classical use of Lévy search for a relocated target [[**CITE**]].

This construction connects to existing frameworks. In reinforcement learning, penalising previously visited states is the basis of count-based and novelty-driven exploration [[**CITE**]], and the kernel can be read as a continuous-time version of such a bonus. In active inference, behaviour minimises an expected free energy with an epistemic part, the information gained, and a pragmatic part, progress towards a preferred state [[**CITE**], 42]. In our framework, the potential V_0 would carry the pragmatic goal and the information gain IG the epistemic part, so their balance could be tuned through the kernel strength and the memory rate γ. The same idea applies to sequence models. Fractional neural attention (FNA) replaces dot-product attention by Lévy diffusion over tokens, so that short- and long-range dependencies arise from a single operator [194]. In this setting a dynamical goal corresponds to a query that changes with the task, and a kernel of previously attended positions could discourage redundant attention in long contexts, with the stability index α made adaptive across layers or heads. As with sampling, these connections are hypotheses, and each would require its own benchmark.

Chapter 5: Discussion

In this thesis we have presented the first model to unify the rich multiscale dynamics of visual search. Previous work has been focused between the two distinct scales of gaze dynamics: the slow timescales and large spatial scales of saccades, or the fast timescales and fine spatial scales of fixation in gaze. In this thesis we unify these scales under a single physics-inspired theory. Further, motivations for previous work on gaze dynamics has also been divided in either aiming to present a biologically accurate understanding of visual search, or a full mechanistic generative model of eye-tracking data. Taking aim to unify these motivations, we constructed a predictive model from the neural dynamics developed under the theory of fractional neural sampling (FNS) [11, 33], and present methods and empirical studies which can fit real human data to the theory.

The methods and dynamics here studied are of importance to physicists as an applied description of a non-equilibrium complex dynamical system. First, visual dynamics are shown to be superdiffusive, and moreover display robust monoscaling, governed by a single stability index α. Such anomalous diffusion at every scale is of interest when studying how basic computational dynamics may be constructed from complex dynamical systems at criticality. We characterised the short-time diffusive scaling of eye-tracking data through fractional absolute moments (Section 3.3), and compared these to the monoscaling superdiffusion of Lévy dynamics (Section 2.4). Such rich multiscale self-similarity provides a physical explanation as the result of control from a neuronal network at criticality with diverging correlation length, resulting in nonlocal spatial access to an informative landscape [195–199].

Second, the model presents and explains the unique pattern of nested oscillations: the slow theta-scale rhythmic transitions between regions of interest, which modulate fast gamma-scale local sampling dynamics, arising from the broad patterns of neural computation, and the confinement of a sampling process. Nested timescales are a recurrent feature of coherent neural dynamics, especially during communication of information between areas of the brain [9, 59, 84–87]. The present model carries this framework over to gaze dynamics, as they emerge from the same neural dynamics which control the process of active vision.

Third, the dynamical-informative kernel introduced in this work describes how visual dynamics are influenced by the history of the gaze trajectory. In particular the informational efficiency of gaze trajectories and the phenomenon of inhibition of return (IoR) suggest a strong dependance on the informative history of a search. Our formulation naturally encodes this information through a belief landscape which evolves from a prior saliency π_0 to a posterior belief after exploration π_t. The cumulative information gathered is then well defined through the Kullback–Leibler (KL–) divergence. Such a well-motivated physics-inspired development of interpretable memory-effects, heavy-tailed noise, and nonlocality is important in the development and progress of physics as an interdisciplinary field.

This work advances the study of the brain in neuroscience, as our model turns gaze into a source of interpretable parameters modulating neural attention, a very important topic of study when understanding decision-making in the brain. The active parameter space here defined through the stability index α, memory–momentum strength β, noise scale γ, and informative timescale t_θ: each parameter has a well-understood and motivated interpretation in neural dynamics. Because gaze aggregates the activity of many brain regions, designing experimental paradigms to fit parameters and summarise collective properties of the circuits that control and process visual input is very impactful to take non-invasive measurement of the brain.

Several experiments may be developed to apply this framework to understand brain health in particular. An experimental protocol tracking high-frequency eye movement alongside coarse-grained saccade positions during a standardised search task without head restraint, combined with head acceleration measurements, would resolve gamma-scale fixational dynamics in naturalistic, head-free viewing and separate eye from head contributions. During such an experiment, coherence of the high-frequency

sampling during fixations can be examined rigorously, as well as testing the saliency-frequency coupling and mapping the paramter-space dynamics as described in Section 3.4. Such an experiment is simple to design using new low-cost and non-invasive techniques such as described in [58] and the data extracted is tractable to analyse across the scales of interest here, to test the predictions of FNS dynamics here presented.

The same sampling process applies to machine learning, in particular to sampling accurately from a given posterior distribution $\pi_{\text{post}}(\boldsymbol{\theta})$ over a set of parameters $\boldsymbol{\theta}$, a problem often handled through the use Markov chain Monte Carlo (MCMC) sampling. Standard MCMC methods are based on sampling a chain of points in parameter space, and proposing local moves through a probability landscape $V(\boldsymbol{\theta}) = -\ln \pi_{\text{post}}(\boldsymbol{\theta})$, often incorporating momentum-coupled Langevin dynamics through parameter space as introduced in Section 1.2. Such dynamics often are unable to robustly explore and mix between well-separated modes, and designing an algorithm to robustly explore a wide range of unseen heavy-tailed or multimodal distributions efficiently without fine-tuning of algorithmic hyperparameters remains a long-standing and active topic in machine learning [143, 144, 200]. The superdiffusive sampling of this thesis address this issue directly, as we have found that in an unseen search task, superdiffusive jumps will efficiently gather informative samples. The momentum-driven local oscillations have also been shown to improve sensitivity to sampled information and efficient sample mixing, as found in similar momentum-driven dynamics in Hamiltonian Monte Carlo [201, 202] and second-order Langevin sampling [202, 203]. Further, the adaptive kernel resembles well-tempered metadynamics [193], in which deposited bias discourages revisiting explored regions, robustly improving ergodicity without reliance on parameter tuning on the unknown landscape $V(\boldsymbol{\theta})$. The link to free-energy minimisation and active inference [42, 152, 193, 204, 205] is important when formalising this link, since both theories consider search as a minimisation of uncertainty about a landscape.

Our results emphasise that coupling oscillations across scales is itself important for an abstract computational sampler: rapid local oscillations explore each mode, while slower rhythmic transitions decide when to move on. Superdiffusive dynamics in generalised computational systems have recently attracted considerable interest, both from theoretical studies of such dynamics emerging from informational sampling [117, 206–208] as well as in designing algorithms in MCMC sampling of abstract distribution landscapes [144, 145, 163, 209, 210]. The same principle of heavy-tailed nonlocality in computation has also influenced more general AI frameworks, notably in work from our group on Fractional Neural Attention (FNA) [194]. FNA is designed to token interactions through Lévy diffusion, as governed by the fractional Laplacian, so that short- and long-range dependencies arise from a single operator rather than hand-designed local–global patterns or fine-tuned scales of influence. In the reported experiments it matches or exceeds Transformer baselines on text classification, image classification, and machine translation, in some cases with a single layer, and its attention networks show larger spectral gaps and shorter path lengths, consistent with faster information mixing. That the same mechanism describes both gaze and attention in artificial networks supports the view that multiscale, nonlocal sampling is a general computational strategy.

In conclusion, this thesis has shown that the rich and apparently disparate dynamics of human gaze can be described by a single physical model with a small set of interpretable components which derive from neural dynamics: nonlocal Lévy jumps, nested oscillations, and an adaptive, information-gathering search landscape. Gaze is both easily measured and serves as a strong proxy for neural decision-making and attention, so gaining an understanding of gaze dynamics gives access to basic computation in the brain. The model offers a quantitative framework making use of concepts from the statistical physics of diffusive processes, makes testable predictions for new experiments to test the theory, and provides a process through which we can understand neural attention at a large scale, and interpret it to develop and understand brain health and basic function of abstract dynamical computational systems.

Appendix A: Prior models of visual search

Model	Core domain	Primary aim	Key features
Ideal Bayesian searcher (IBS) Najemnik and Geisler [37], see also [35, 36, 155]	Information- and decision theory	Establish an optimally informative process for saccade localisation based on biologically realistic information foraging of the eye.	Exact Bayesian belief updating, incorporating biologically realistic retinal resolution gradients, spatial noise, and global entropy reduction. Considers discrete saccades rather than continuous time.
Least action / active inference Zanca and Gori [42]	Machine learning	Generating scanpaths (trajectories) through physics-inspired variational informational free-energy minimisation.	Generative deterministic model of scanpaths. Sophisticated account of natural image saliency. Does not consider fixational dynamics, does not aim to be biologically realistic.
Self-avoiding random walk Engbert et al. [7], see also [77]	Statistical physics and oculomotor control	Develop an accurate account of low-frequency ocular drifts during free-viewing fixations.	Lattice-based stochastic motion with self-avoidance constraints; 2D quadratic potential. Does not capture a high-frequency tremor or saccades.
DeepGaze Abdelrahman, Al-Tawil, and Al-Hamadi [211], see also [47, 48, 212, 213]	Deep learning and computer vision	Construct a neural network to predict human scanpaths on natural images with high empirical accuracy.	Modular deep neural network trained on eye-tracking data; models internal state, memory, and information. Not designed as a biologically realistic model; considers discrete saccade locations.
Itti–Koch–Niebur model Itti, Koch, and Niebur [74]	Biologically inspired vision	Simulate bottom-up, stimulus-driven visual attention based on early primate visual pathways.	Multi-scale feature extraction (colour, intensity, orientation); winner-take-all selection with inhibition of return (IOR).
Feature-integration theory Gayet and Aydemir [76]	Cognitive psychology	Explain experimental differences in performance during sparse- and dense-search tasks.	Pre-attentive direction of attention based on saliency theory. Biologically motivated, but not generative or explanatory for eye movement measurement.
Guided search Wolfe [75] and Wolfe, Cave, and Franzel [214]	Cognitive psychology	Account for the continuum of multi-scale search efficiency by combining bottom-up and top-down guidance of attention.	Parallel feature maps weighted by stimulus salience and target template to form an activation map; serial deployment of attention in decreasing activation order.

Table A.1: Review of a selection of influential models of visual search and gaze dynamics. Comparison across core theoretical domains, primary objectives, and defining operational features. Models span a spectrum from biologically grounded frameworks prioritising physiological mechanisms (e.g., Feature-Integration Theory, Itti–Koch–Niebur) to normative and computational architectures optimised for empirical trajectory prediction on complex natural scenes (e.g., Ideal Bayesian Searcher, DeepGaze). Important features which share analogy in the model presented in this thesis include: an information kernel (in discrete time) from the approach in Najemnik and Geisler [37], the second-order (noiseless) dynamical system coupled to a rich account of saliency gradients presented in Zanca and Gori [42], and the description of the rich non-tremor fixational random motion in the presented in Engbert et al. [7].

Appendix B: Code availability

Repository	Function
VisualFNS	Producing simulations, visualisations, and raw data of gaze trajectories through numerically integrating the system Equation (2.3) with kernel adaptation under Equation (2.5). Important techniques in solving this system were the use of the Euler–Maruyama numerical scheme for SDE integration and the Fourier-space spectral solution of kernel updates.
VisualFNS_Analysis	Analysing trajectory data which was output from VisualFNS. Involved producing power spectra, marking intervals between saccade-like transitions, and analysing superdiffusive scaling.
VisualData_Analysis	Loading in the data provided from Näher et al. [61], and analysing it in the same way as in VisualFNS_Analysis. Involved producing power spectra, marking intervals between saccade-like transitions, and analysing superdiffusive scaling. Originally forked from FractionalSaccades.jl.
FractionalNeuralSampling.jl	The project which provides the computational basis to formalise the FNS equation and its variants in syntax solvable through DifferentialEquations.jl.

Table B.1: A list of repositories containing the code used in this thesis. All code was written in Julia, apart from some helper scripts written in Python.

Appendix C: Eye-tracking supplements

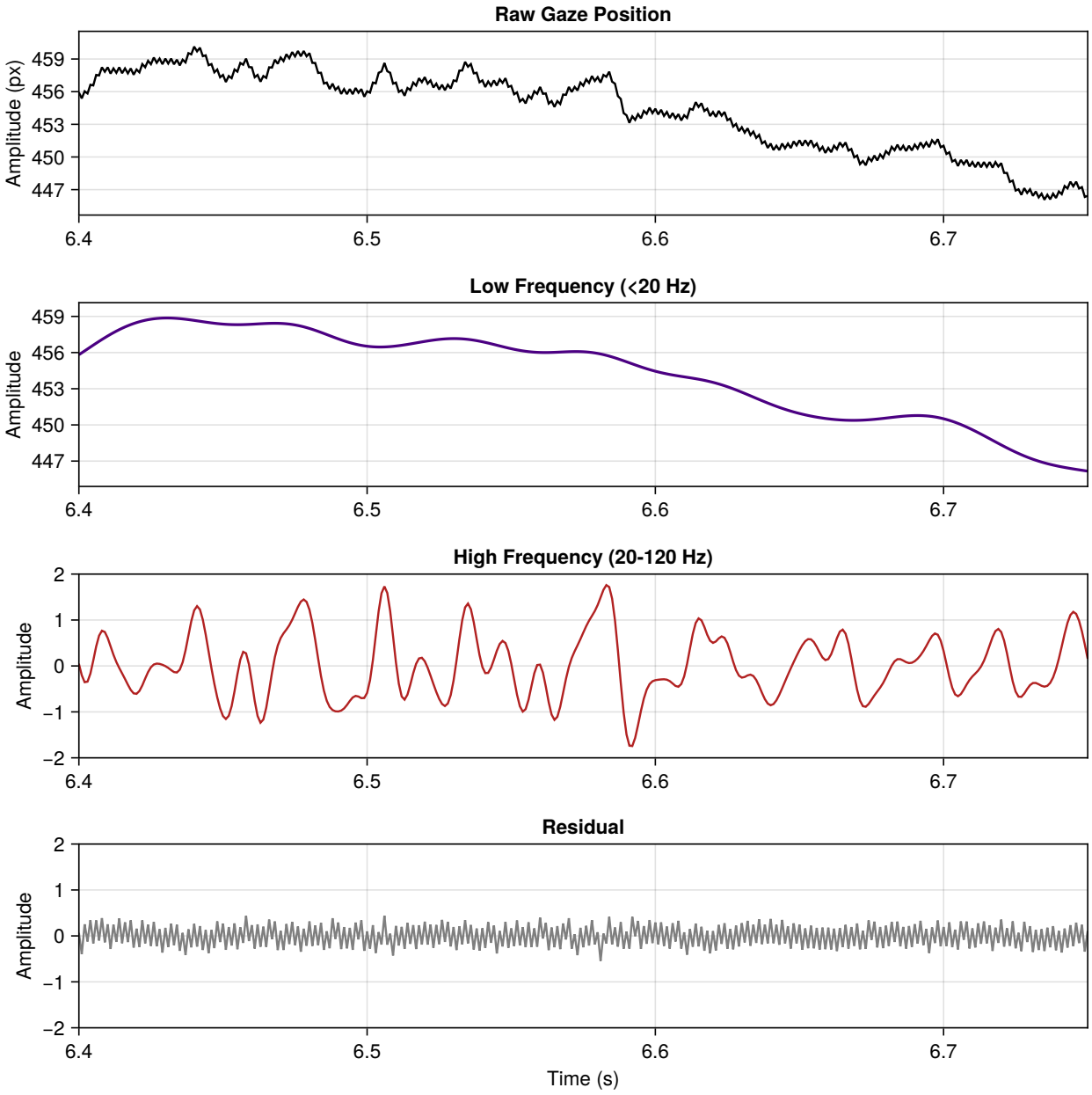


Figure C.1: The Fourier decomposition of a typical fixation.

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